

Ultra-Low-Power PMU

IEEE • SI-CLASH

CMOS 65nm technology

Band Gap Reference

1 Weekly Objective

This update documents the literature study and the planned transistor-level design methodology for the Bandgap Reference (BGR) block. The selected starting point is the supplied 2021 Electricity paper, which presents a low-voltage, low-power reverse/current-mode bandgap implemented in 65-nm CMOS. No claim is made here that the SiClash BGR has already been designed or simulated.

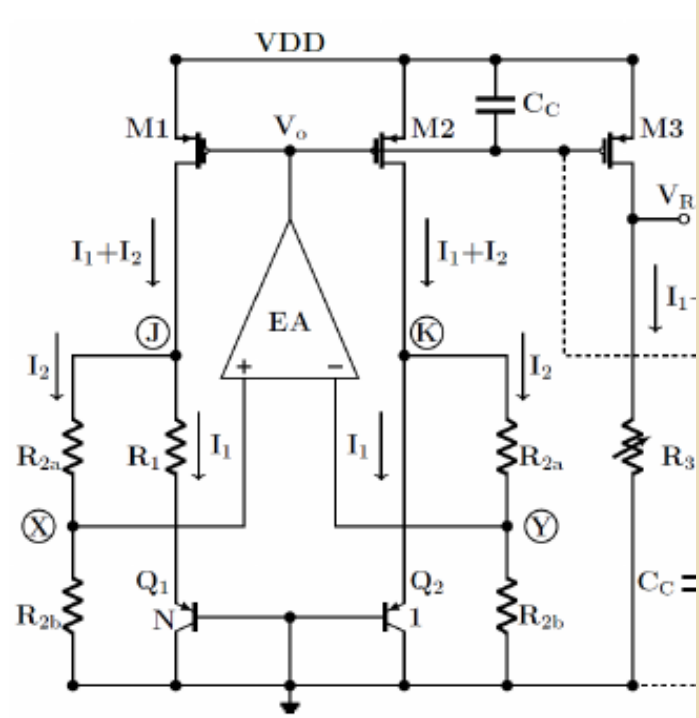
The design will use the paper's topology and equations as a starting methodology, then re-derive the resistor ratios, current levels, device sizing, amplifier requirements, startup behavior, and verification plan for the SiClash specifications. The paper's numerical component values are reference examples only and will not be copied.

2 SiClash BGR Requirements

Requirement	Target / Interpretation
VBG target	1.000 V nominal; allowed range 0.995–1.005 V
Temperature	5 °C to 45 °C; typical 31 °C
Supply context	PMU analog VDD: 1.1–1.3 V, nominal 1.2 V
Accuracy	±0.5% (±5 mV)
Bias output	100 nA bias current is required by the PMU; this is a separate Bias Generator output, not the BGR voltage output
Startup	≤1 ms to 0.5% accuracy
PSRR	≥10 dB from 1 Hz to 1 MHz
Control	Enable input; disabled state is power-down
VBG_OK	Assert when VBG reaches ≥95% of steady-state value
DFT	Sense VBG and bias current; force VBG (and, at PMU level, bias current) through ATB
Noise	1 Hz–10 kHz integrated noise plus 1-Hz and 1-kHz spot-noise reporting

Source: SiClash project problem statement, especially R.013–R.023 and the DFT requirements.

3 Selected Literature Architecture



Current-mode / low-voltage reverse bandgap architecture reproduced from the supplied paper.

The paper separates the circuit into the bandgap branches and an error amplifier (EA). The EA forces the two sensing nodes toward equality, while the resistor network converts the CTAT VEB term and PTAT ΔVEB term into the reference. The paper emphasizes accuracy over bandwidth and recommends larger devices / longer channel lengths when gain and matching are more important than speed.

4 Design Equations and How They Will Be Used

PTAT voltage: $\Delta VEB = VT \ln(N)$, where $VT = kT/q$

Bandgap design relation: $R1 = \Delta VEB / I1 = VT \ln(N) / I1$

Temperature compensation: $R2 = m\ R1 / \ln(N)$, where m is technology-dependent and evaluated near the nominal temperature

Output scaling: $VREF = (1/\gamma)[(R3/R2)VEB + (R3/R1)VT \ln(N)]$

Error model: $VERR = Voff(R3/R1)(1 + R2a/R2b) + \varepsilon R \cdot VREF + \varepsilon M \cdot VREF$

Minimum BG supply: $VDDmin,BG = VX + |VGS,sth| + 2VDS,sat$ (for the cascode implementation discussed in the paper)

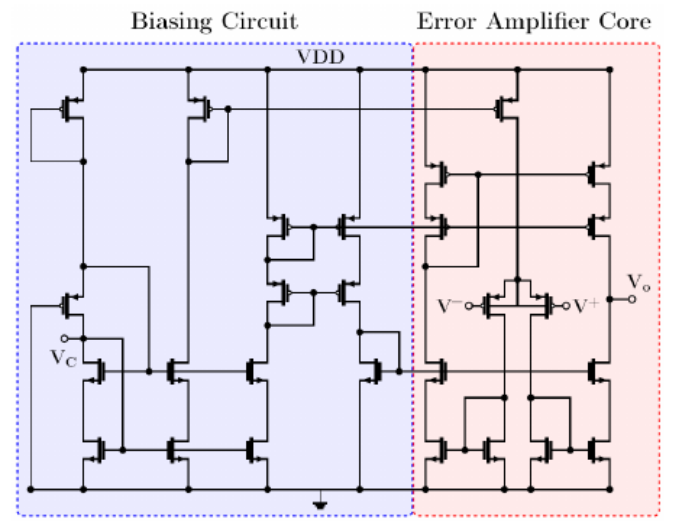
DC PSR: $PSRDC = [R3(1 + gm,M3r0)/(R3+r0)] \cdot 1/[1 + gm,ea \cdot rea \cdot gm,M3(RJ-RK)]$

PSR compensation: $z1c = (1 + gm,M3r0)/(rea\ CC)$ when CC is connected from Vo to VDD

The first design pass will determine N and the branch current from the required PTAT signal and available power. R1 is then obtained from ΔVEB/I1. R2 follows from the technology-dependent temperature coefficient parameter m. Finally R3 and γ are selected so that the nominal reference is 1.000 V rather than the 600 mV target used in the paper.

The temperature-compensation target will be verified around the SiClash nominal temperature of 31 °C, followed by a full 5–45 °C sweep. The resulting curvature, rather than only the first-order slope, must remain inside the ±5 mV accuracy window.

5 Error Amplifier Strategy

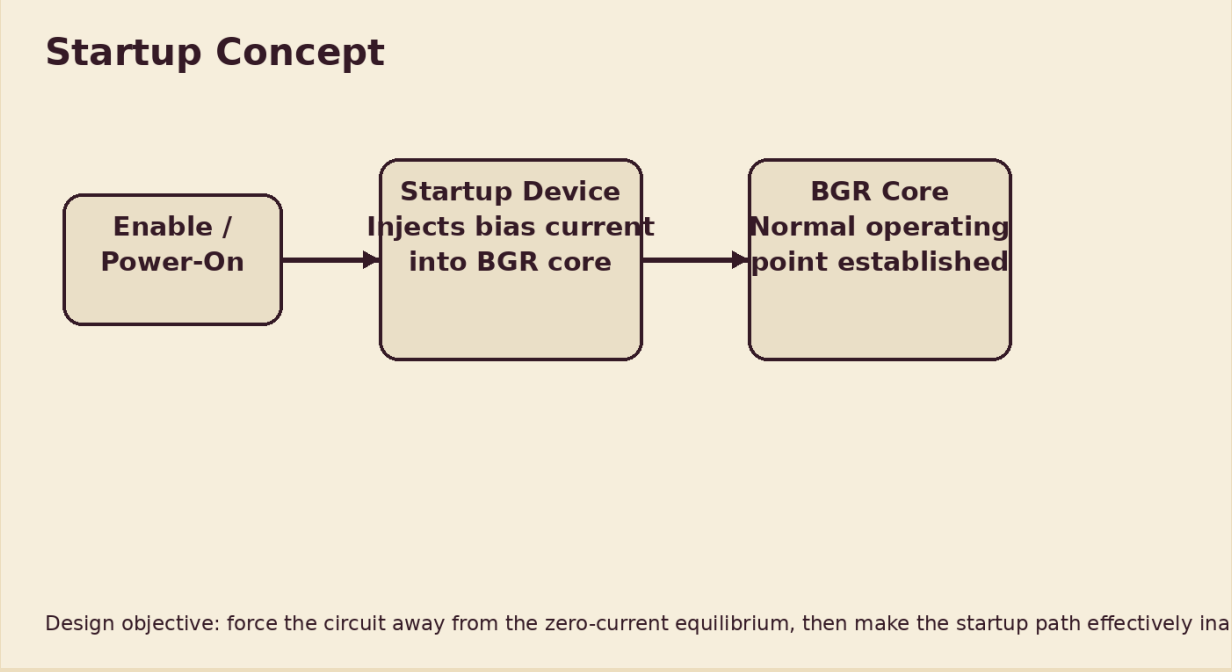


Error-amplifier/biasing implementation shown in the supplied 65-nm paper.

The paper treats the EA as a static-accuracy block: DC gain, input offset, and biasing are more important than large bandwidth. It discusses symmetrical, two-stage Miller, and folded-cascode options. For the SiClash design, the EA topology will be selected after the BGR headroom and current budget are known.

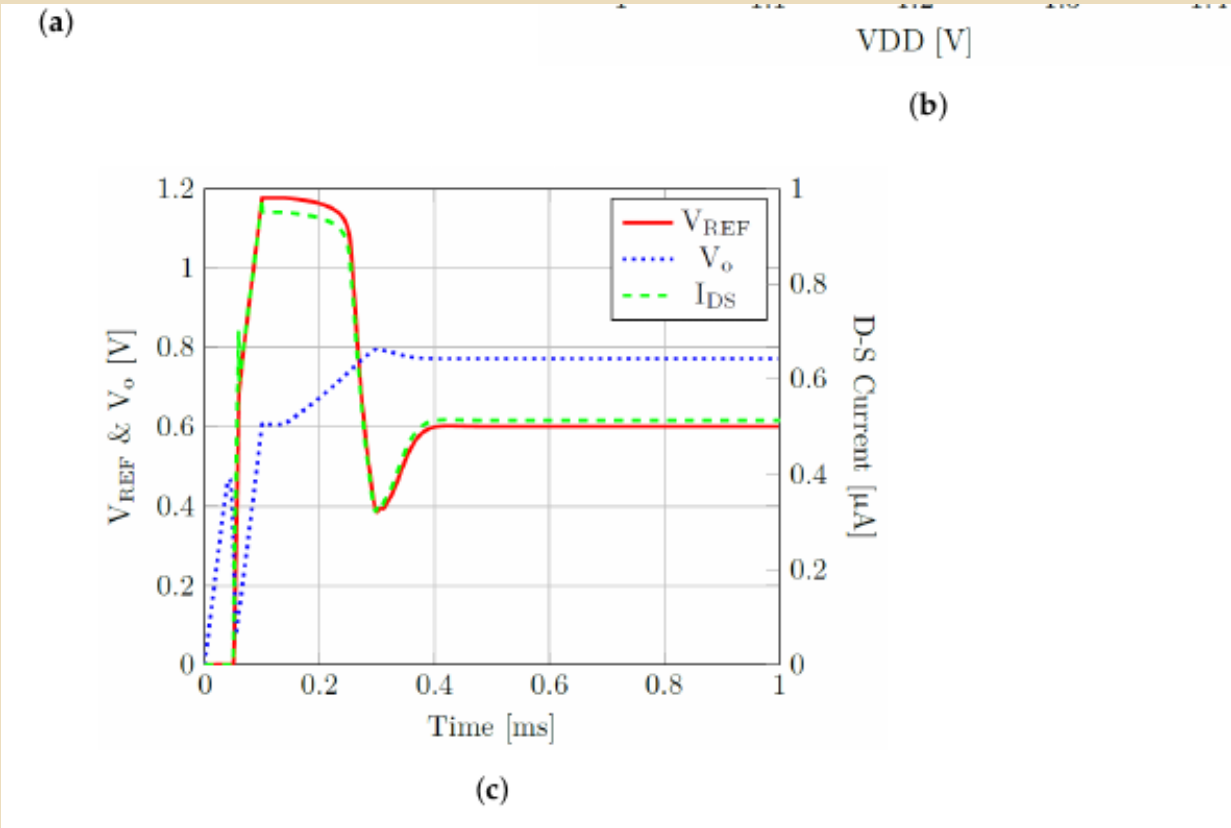
A useful initial design target is high DC gain with low offset, but the final number will be derived from the ±5 mV system error budget and verified across PVT/Monte Carlo. The paper's ~70–80 dB values are not adopted as a project requirement; they are a reference point for the design study.

6 Proposed Startup Implementation



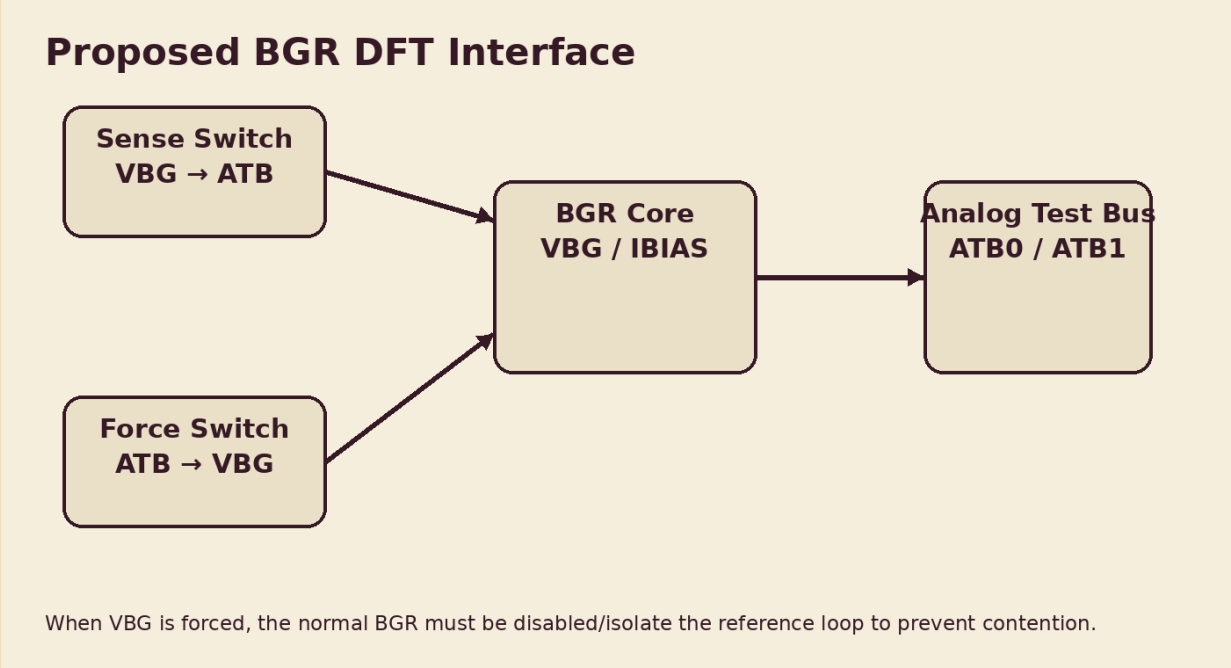
Planned startup function for the SiClash BGR.

The supplied paper uses a startup transistor Msu to inject current into the BGR when an unwanted zero-current operating point is possible. Once the normal operating point is reached, Msu turns off. The SiClash implementation will use the same functional principle but will be sized and controlled for the project's ≤1 ms startup requirement.



Startup transient behavior reported in the supplied paper; shown as a reference waveform, not as a SiClash result.

7 DFT: Sense / Force Plan



Proposed DFT structure derived from the SiClash sense/force requirements.

For sense mode, a test switch connects the internal VBG to an Analog Test Bus (ATB). The switch must have sufficiently low leakage and controlled ON resistance so that the normal BGR is not materially loaded. For force mode, an external ATB voltage overrides VBG while the normal BGR loop is disabled or isolated. The project explicitly requires that the BGR be disabled when VBG is forced.

At PMU level, the supplied specification also calls for sensing and forcing the 100-nA bias current. That current belongs to the separate Bias Generator block, so its DFT switch will be implemented with the same isolation philosophy.

8 65-nm Redesign Workflow

1. Confirm the available PDK device options, especially whether the process provides a suitable PNP/BJT device model and what operating region is valid.
2. Select N and a nominal branch current using the required ΔVEB, power budget, mismatch, and headroom trade-offs.
3. Extract/estimate the technology-dependent VEB(T) behavior and parameter m around 31 °C from the PDK simulations.
4. Calculate R1, R2, R2a/R2b, R3 and the mirror ratio γ from the equations above for a 1.000-V target.
5. Implement the bandgap core first, without DFT or startup, and verify DC operating point and temperature compensation.
6. Design the EA for sufficient DC gain, low offset, correct headroom, and acceptable current.
7. Add startup and verify ≤1 ms settling to 0.5%.
8. Add DFT sense/force switches and verify that forcing VBG disables the normal BGR.
9. Run supply, temperature, PVT, Monte Carlo, PSRR, and noise simulations and build the full error budget.

9 Verification Plan — To Be Filled With Cadence Results

Test	Condition	Pass Criterion	Status
VBG DC	TT, 31 °C, VDD=1.2 V	0.995–1.005 V	Planned
Temperature	5–45 °C	Within ±5 mV and acceptable TC	Planned
Supply	VDD=1.1–1.3 V	PSRR / line sensitivity meets allocation	Planned
PSRR	1 Hz–1 MHz	Minimum ≥10 dB	Planned
Noise	1 Hz–10 kHz	Report integrated + 1 Hz + 1 kHz	Planned
Startup	Enable 0→1	≤1 ms to 0.5%	Planned
Powerdown	Enable 1→0	BGR disabled; leakage/battery budget respected	Planned
VBG_OK	Startup transient	Assert at ≥95% VBG	Planned
DFT sense	TM sense enabled	ATB observes VBG without unacceptable loading	Planned
DFT force	TM force enabled	VBG overridden; BGR disabled	Planned
PVT	TT/FF/SS + temperature/supply	Functional and within allocated error	Planned
Monte Carlo	Mismatch + process	Distribution fits accuracy budget	Planned

10 What Has Been Completed This Week

• Reviewed the supplied SiClash PMU requirements relevant to the BGR, including 1-V accuracy, temperature range, PSRR, noise, startup, VBG_OK, enable/powerdown, and DFT.

• Studied the supplied 65-nm low-voltage reverse/current-mode bandgap paper and identified the bandgap branches, EA, current-mirror strategy, startup, and PSR-compensation mechanisms.

• Selected the paper's methodology as the starting point while explicitly avoiding direct reuse of its 600-mV target and transistor/resistor dimensions.

• Defined the equations and a transistor-level implementation workflow for the next design stage.

No Cadence schematic, sizing, simulation result, PVT result, or Monte Carlo result is claimed in this weekly update.

11 Next Design Actions

• Inspect the 65-nm PDK device library and identify the valid BJT/PNP or alternative reference devices.

• Characterize the chosen device(s) in Cadence at the SiClash temperature and supply conditions.

• Choose N and initial branch current; calculate resistor ratios and first-pass resistor values.

• Implement the BGR core and establish the correct DC operating point before adding startup/DFT.

• Design the EA around the required accuracy and headroom, then run the first temperature and supply sweeps.

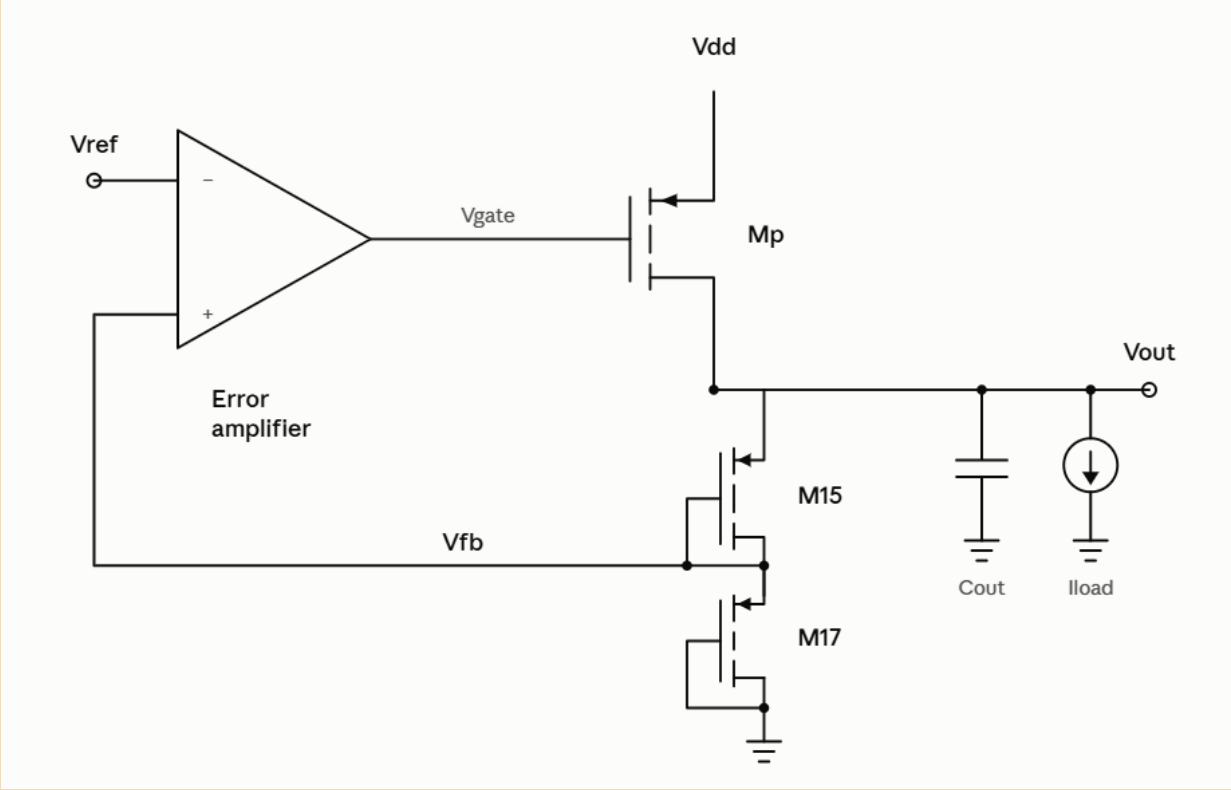
12 References

Barteselli, E., Sant, L., Gaggl, R., & Baschirotto, A. (2021). Design Techniques for Low-Power and Low-Voltage Bandgaps. Electricity, 2, 271–284. DOI: 10.3390/electricity2030016. Supplied project reference.

SiClash / Ultra-Low-Power PMU for a Biomedical Sensor Interface — supplied project problem statement, 2026.

Figures reproduced/adapted in this report are taken from the supplied paper for project-study purposes and are identified as such in their captions. The proposed startup and DFT diagrams are original design-planning figures.

LDO



1. Goal this week

Choose an LDO topology for the analog supply VDD (1.2 V, range 1.1–1.3 V, load 1–50 µA) that has very low quiescent current (Iq) and small area, and that can be stable without a large external capacitor.

2. Why Iq matters (budget derivation)

- Battery: 150 mAh, must last 90 days (2160 h), so the average battery current must be ≤ 150 mAh / 2160 h ≈ 69 µA.
- The load alone can take up to 50 µA, which leaves about 19 µA for everything else: bandgap, bias generator, POR, and LDO Iq.
- So the LDO Iq must be only a few µA, and the feedback network must draw much less than that.

3. Topology research findings

The conventional feedback network is a resistor divider. For ultra-low power it has two problems:

- It needs very large resistance (hundreds of kΩ) to keep the current at a few µA, which means large silicon area.
- Process variation of the resistors causes inaccuracy and mismatch, which appears as output DC offset.

4.Chosen approach: replace the resistor divider with a string of diode-connected PMOS transistors biased in subthreshold.

To reduce the quiescent current and avoid the large area associated with conventional high-value resistors, I investigated a feedback network based on diode-connected MOS transistors operating in the subthreshold region.

The basic idea is: Instead of using large passive resistors, diode-connected MOS transistors can provide a high effective resistance while carrying only a small current [1].

The subthreshold region is particularly attractive for ultra-low-power designs because a MOS transistor can operate with very small drain current while still providing the required voltage drop.

5. Advantages of the MOS Feedback Approach

The literature indicates that using MOS devices in the subthreshold region can be attractive for low-power LDOs because:

Low quiescent current: The feedback network can operate with only a few microamperes or less, reducing the overall LDO quiescent current.

Smaller area: A conventional resistor with several hundred kΩ can require considerable silicon area depending on the available resistor technology.

Using MOS devices can provide a large effective resistance to a potentially smaller area.

Suitable for ultra-low-power applications: This approach is particularly useful when the LDO must operate from a limited energy source, and the feedback current must be minimized.

6. Design decisions so far

- **Pass device:** PMOS (Mp), supplied from Vbat. With Vbat ≥ 2.3 V and VDD = 1.2 V there is plenty of headroom, so dropout is not an issue.
- **Feedback network:** diode-connected MOS string, total equivalent resistance ≈ 600 kΩ, drawing about 1.6 µA. This sets Vfb to match Vref (1 V from the bandgap).
- **Consequence:** the feedback node Vfb is now high impedance, which affects the error amplifier and stability.
- **Error amplifier: telescopic OTA**, to get high gain at low current. Since it runs from Vbat (2.3–3.2 V), the stacked devices have enough headroom.

7.Work Completed This Week

1. **Literature Survey:** Identified subthreshold MOS feedback structures to drastically cut passive area while setting divider current to $I_{\text{div}} \leq 1.6\ \mu\text{A}$.
2. **First-Pass Sizing & Feasibility:** Defined operating current targets: $I_{\text{OTA}} \approx 2\ \mu\text{A}$, $I_{\text{FB}} \approx 1.6\ \mu\text{A}$, yielding total $I_Q < 5\ \mu\text{A}$ (well within PMU budget).
3. **Core Schematic Capture:** Built the baseline schematic with a PMOS pass transistor (M_p), subthreshold feedback pair (M_{15}, M_{17}), and single-stage amplifier.

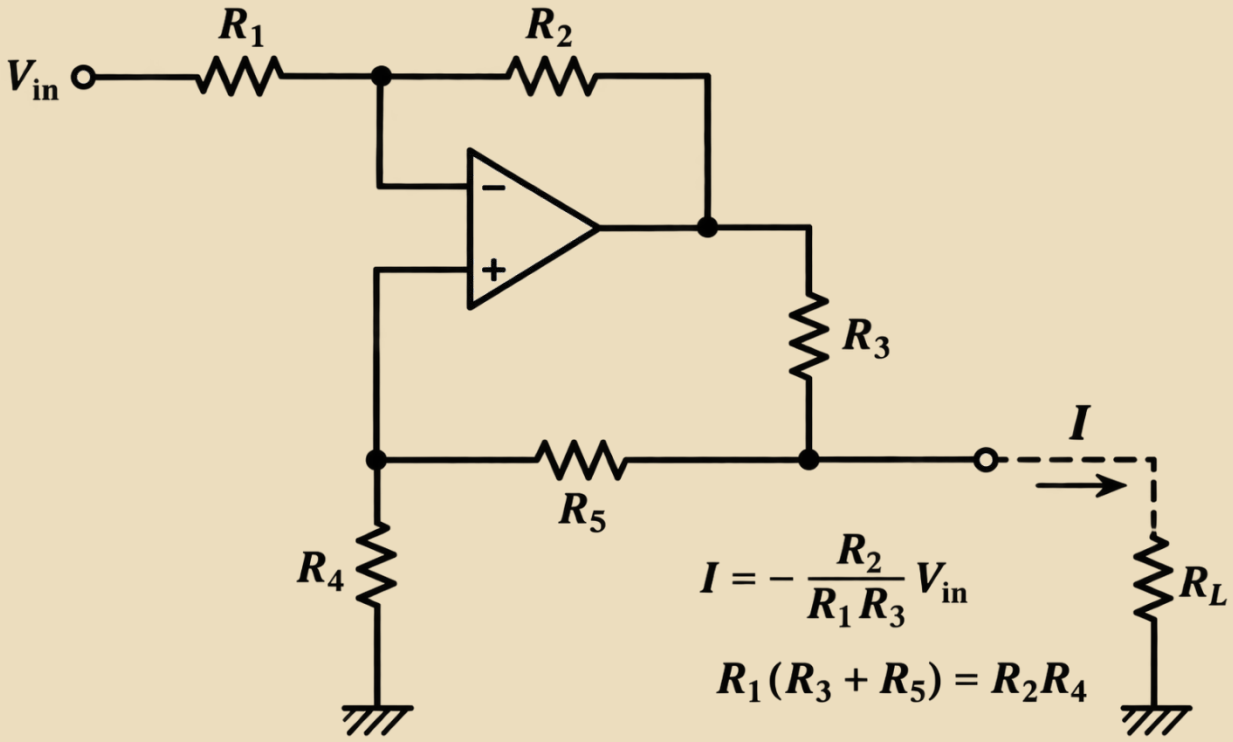
8.References

[1] C. J. Cheng, C. J. Wu, and S. Y. Lee, “A programmable pacing channel with a fully on-chip regulator for cardiac pacemaker,” in Proc. Asian Solid-State Circuits Conf. (A-SSCC), Nov. 3–5, 2008, pp. 285–288.

Bias Generator

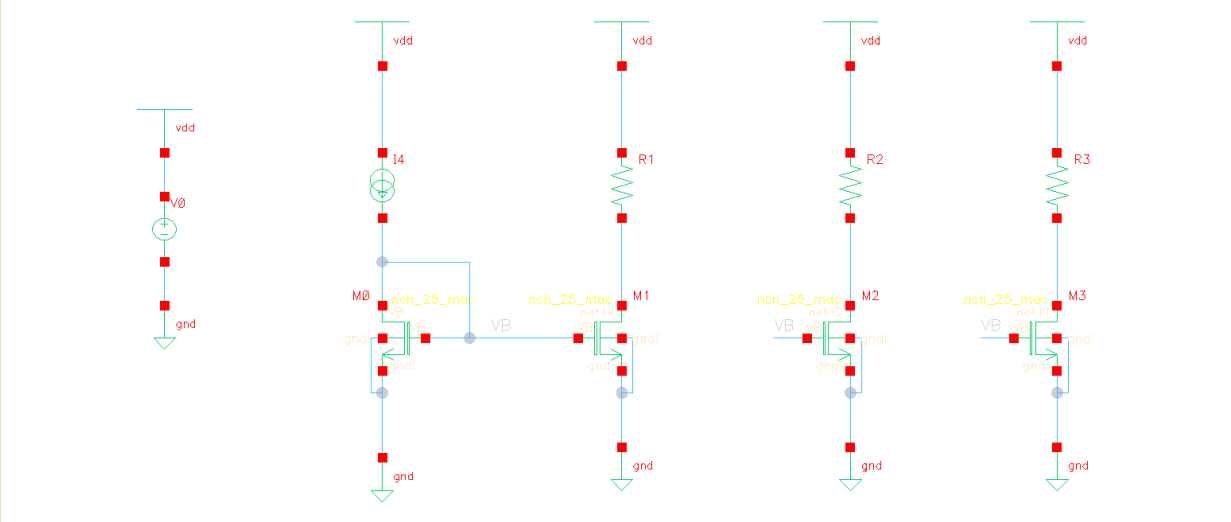
Our input is $1V$ from the BGR, $V_{bat} = 3V$ (Typically), we need to provide stale $100nA$ across the corners and PVT

There is many topologies we can use, for now let's try the "voltage to current converter op-amp topology", and then we can try other topologies.



We can find a way to mirror the output current to minimize its variation.

Current mirror test bench



Sizing & test bench design

For now, let $V_{DD} = 1V$, $g_m/I_D = 15$, $V_{DS} = 0.5V$, max $L = 5\mu m$

Forcing $100nA$ from the DC source with temp coefficient $\alpha = 100 \times 10^{-6}$

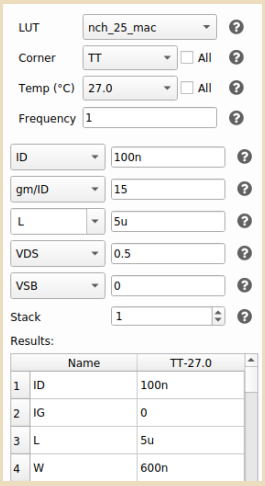
From ADT, we find $W = 0.6\mu m$ to achieve $100nA$ drain current.

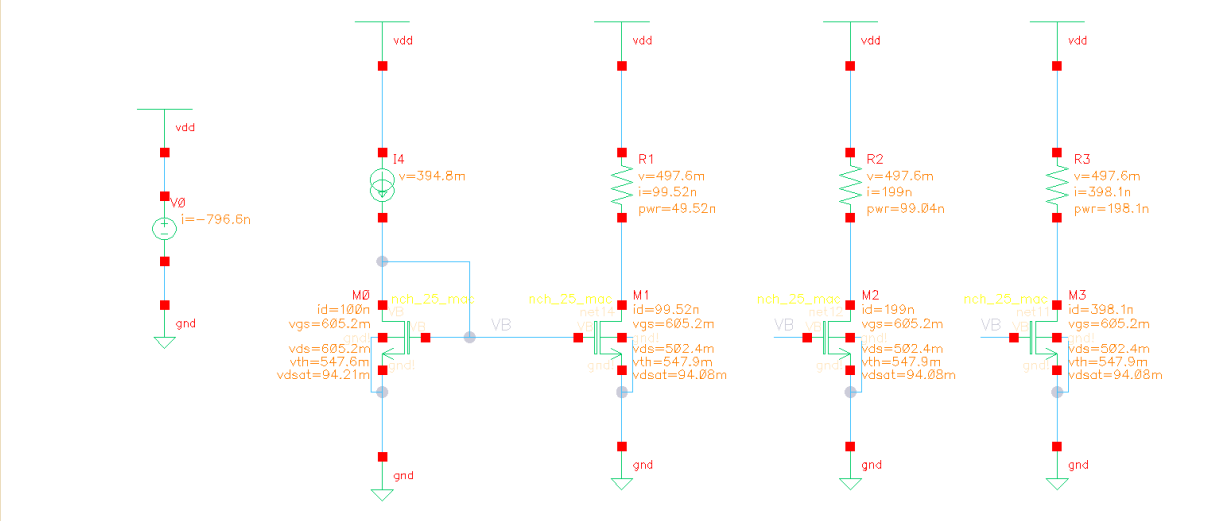
Setting mirroring ratios to get $100nA, 200nA$ & $400nA$

$\Rightarrow m_0 = m_1 = 1, m_2 = 2, m_3 = 4$

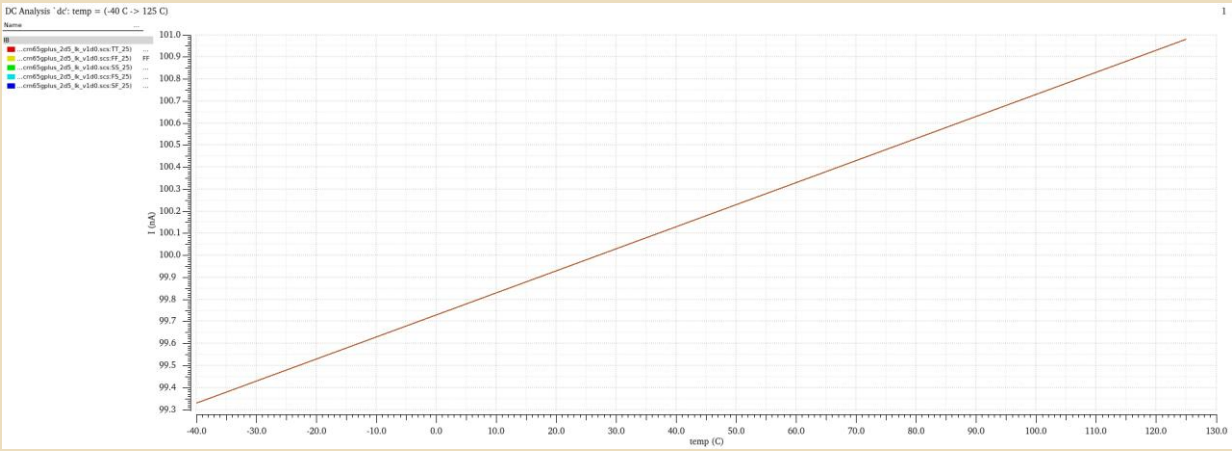
$R_3 = 0.5R_2 = 0.25R_1$.

$R_1 = 0.5V/100nA = 5M\Omega \Rightarrow R_2 = 2.5M\Omega, R_3 = 1.25M\Omega$

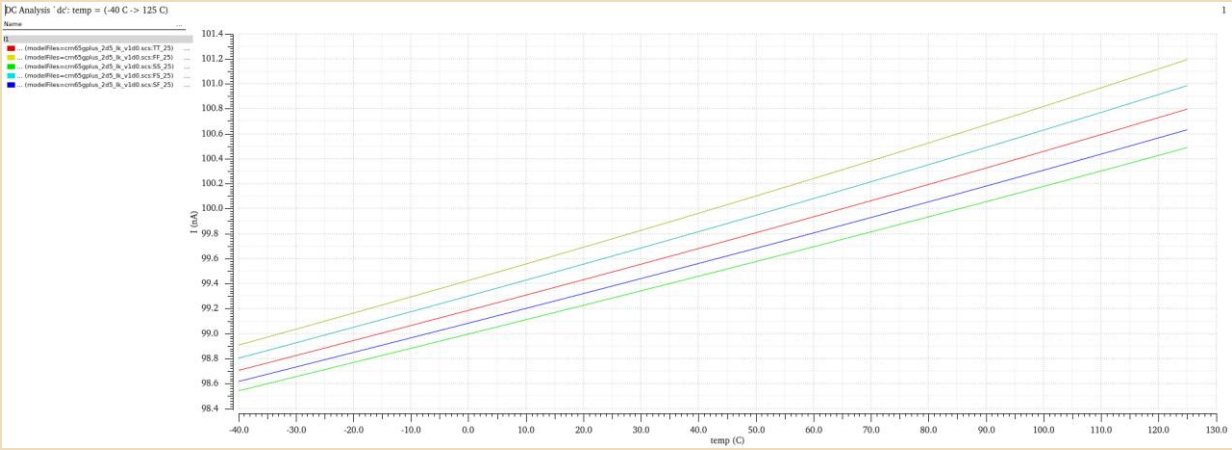




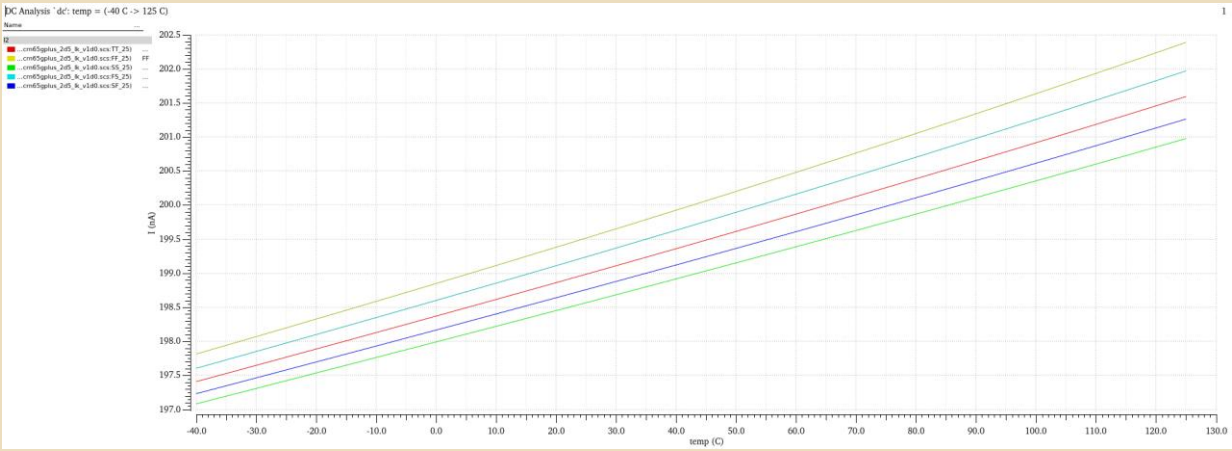
OP simulation



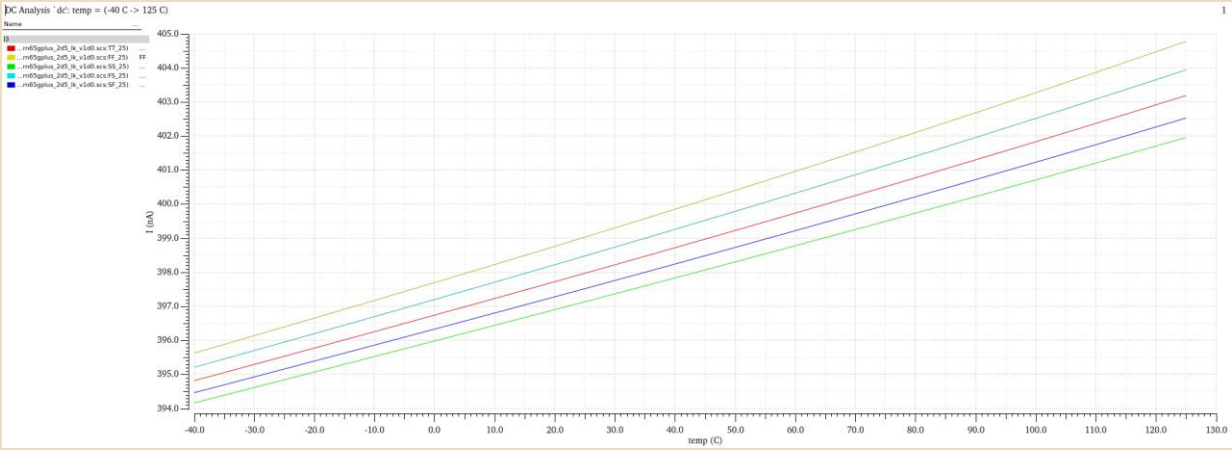
I_B variations



I_1 variations



I_2 variations



I_3 variations

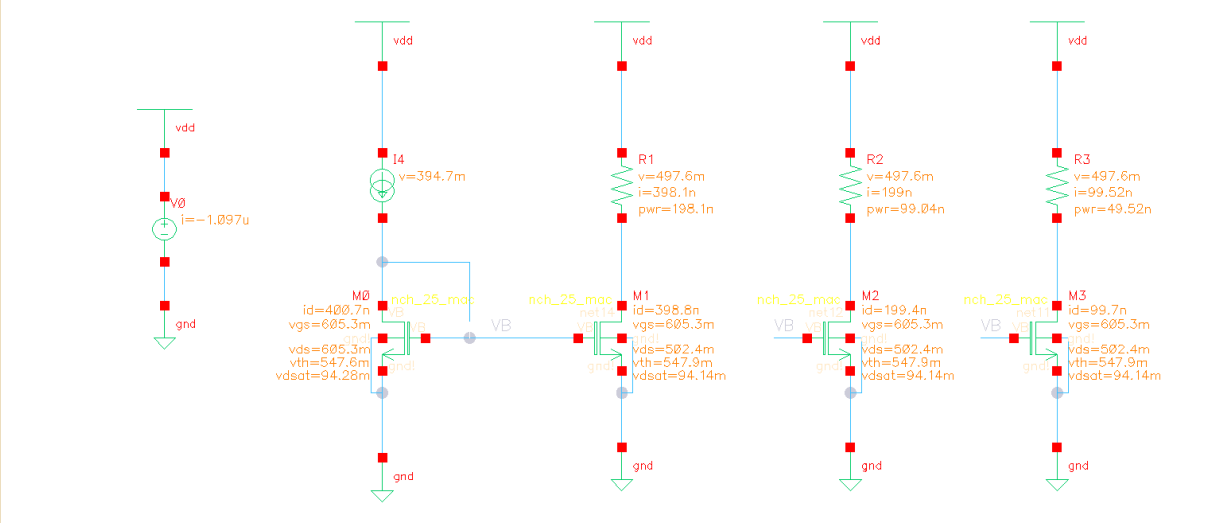
Test	Output	Spec	Weight	Pass/Fail	Min	Max	TT	FF	SS	FS	SF
si-Bgenerator:1-CM-test:1	IB										
si-Bgenerator:1-CM-test:1	IB-variations				1.65n	1.65n	1.65n	1.65n	1.65n	1.65n	1.65n
si-Bgenerator:1-CM-test:1	I1										
si-Bgenerator:1-CM-test:1	I1-variations				1.946n	2.285n	2.091n	2.285n	1.946n	2.181n	2.015n
si-Bgenerator:1-CM-test:1	I2										
si-Bgenerator:1-CM-test:1	I2-variations				3.892n	4.571n	4.182n	4.571n	3.892n	4.362n	4.029n
si-Bgenerator:1-CM-test:1	I3										
si-Bgenerator:1-CM-test:1	I3-variations				7.785n	9.142n	8.365n	9.142n	7.785n	8.725n	8.058n

The ideal expected nominal variation for $I_3 = 4 \times 1.65nA = 6.6nA < 8.365nA$

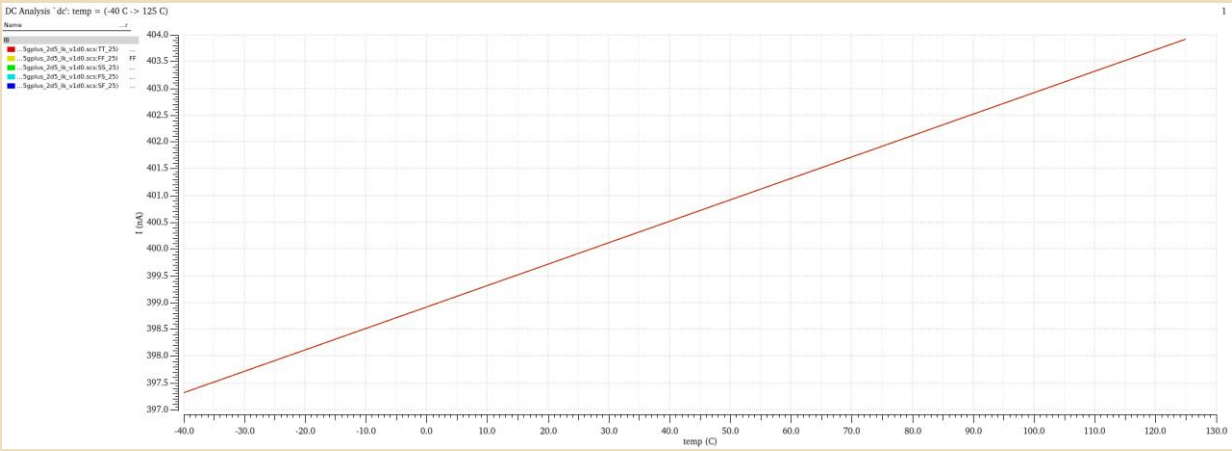
Invert the mirroring ratios to be $m_0 = m_1 = 4, m_2 = 2, m_3 = 1$

Forcing $400nA$ from the DC source with temp coefficient $\alpha = 100 \times 10^{-6}$

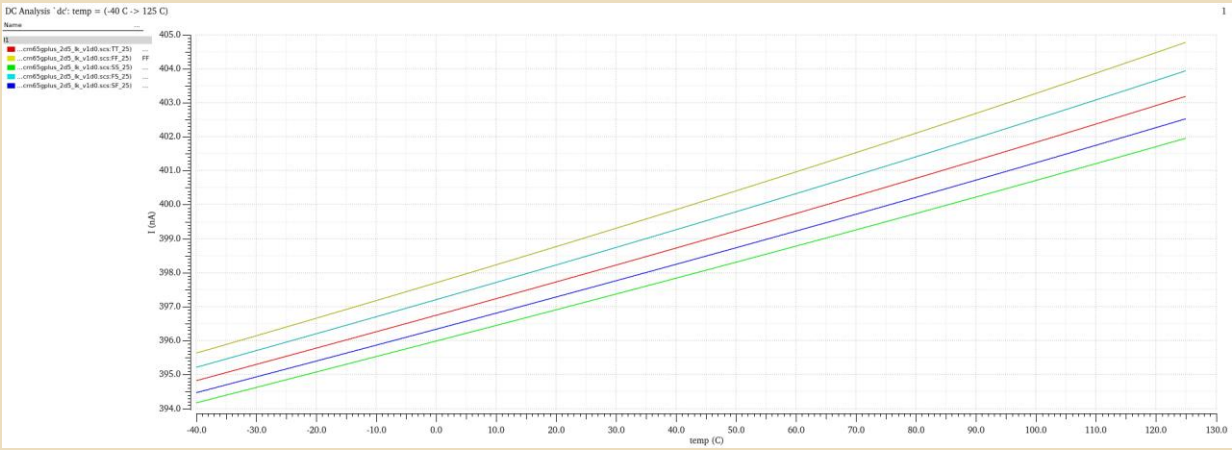
$R_1 = 0.5R_2 = 0.25R_3 \Rightarrow R_3 = 0.5V/100nA = 5M\Omega \Rightarrow R_2 = 2.5M\Omega, R_1 = 1.25M\Omega$



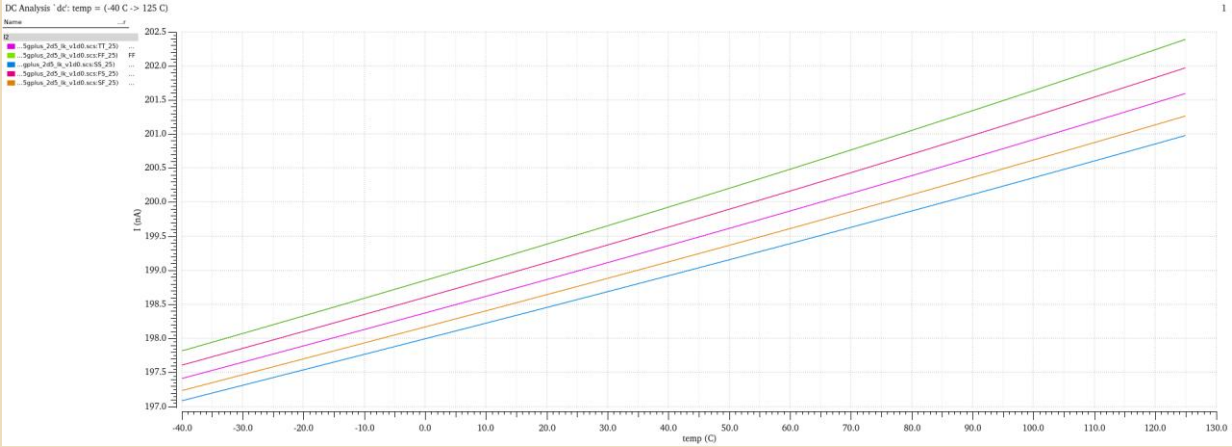
OP simulation



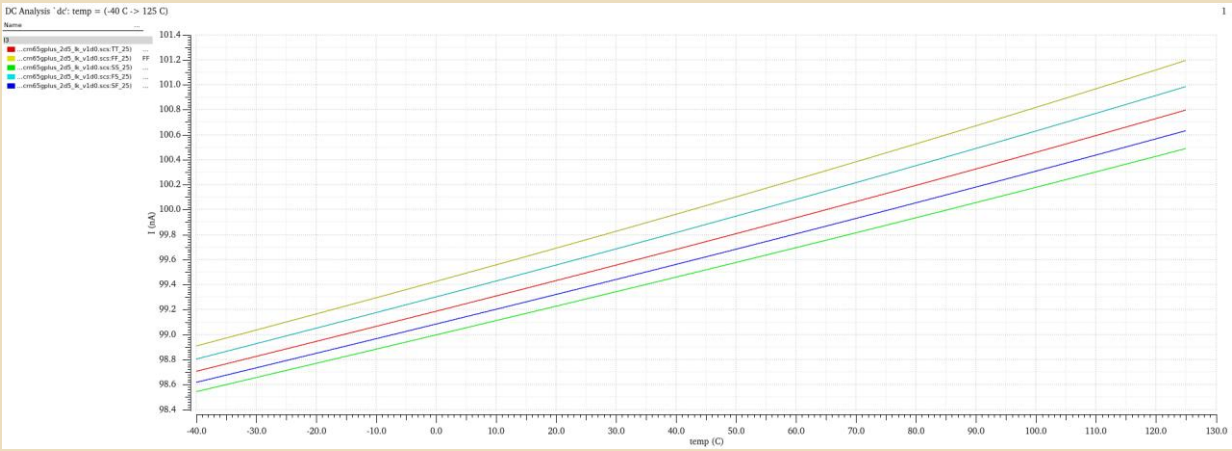
I_B variations



I_1 variations



I_2 variations

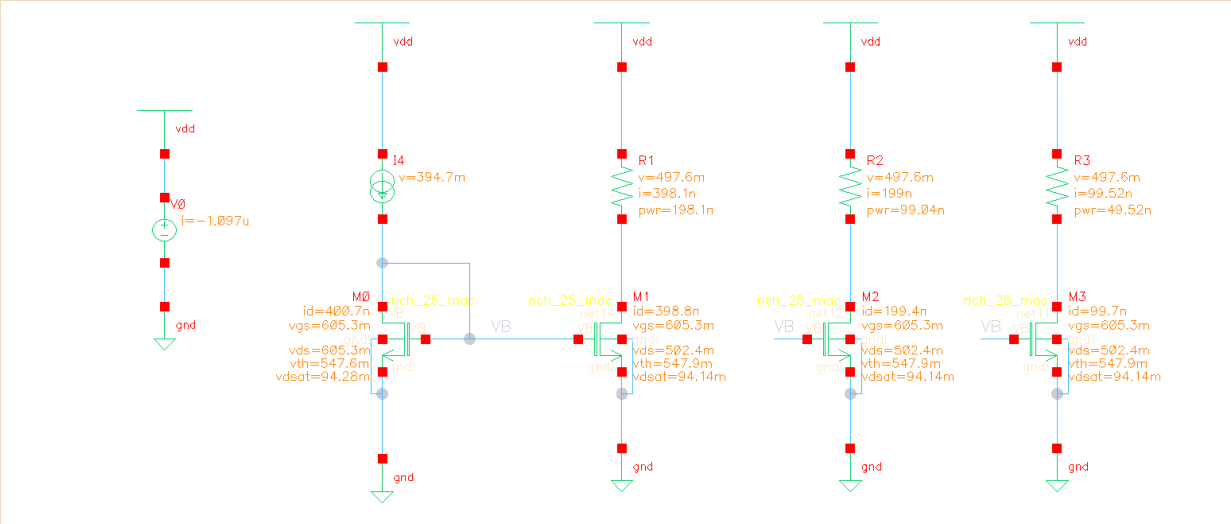


I_3 variations

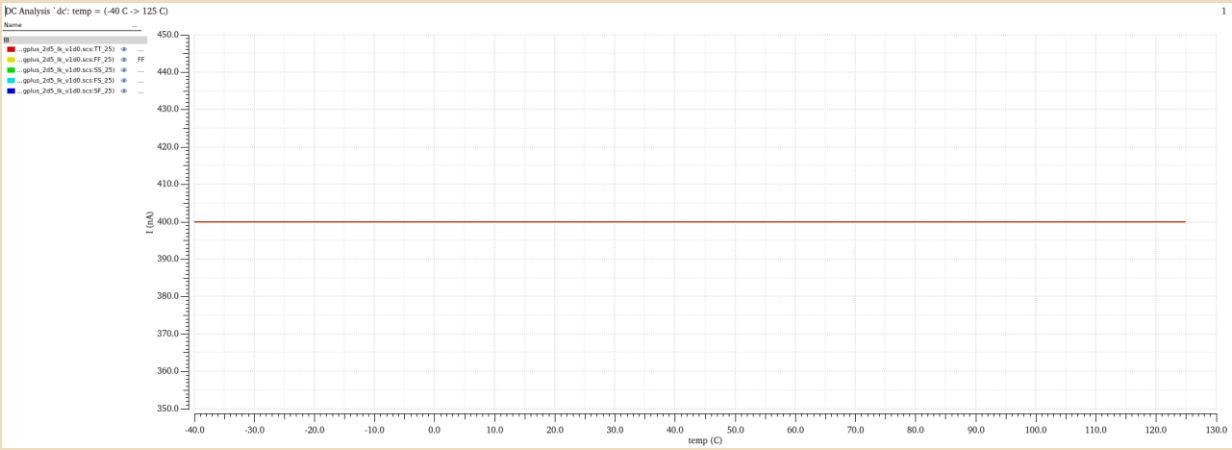
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si-Bgenerator:1-CM-test:1	IB-variations				6.6n	6.6n	6.6n	6.6n	6.6n	6.6n	6.6n
si-Bgenerator:1-CM-test:1	I1										
si-Bgenerator:1-CM-test:1	I1-variations				7.784n	9.142n	8.364n	9.142n	7.784n	8.724n	8.057n
si-Bgenerator:1-CM-test:1	I2										
si-Bgenerator:1-CM-test:1	I2-variations				3.892n	4.571n	4.182n	4.571n	3.892n	4.362n	4.029n
si-Bgenerator:1-CM-test:1	I3										
si-Bgenerator:1-CM-test:1	I3-variations				1.946n	2.285n	2.091n	2.285n	1.946n	2.181n	2.014n

The ideal expected nominal variation for $I_3 = 1.65nA/4 = 1.65nA < 2.091nA$

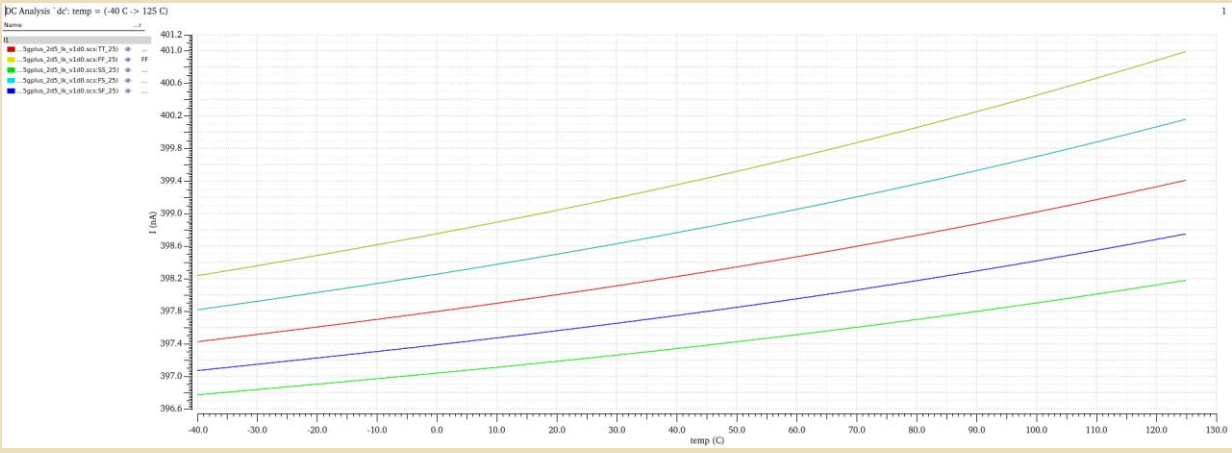
For ideal input, setting the temperature coefficeint $\alpha = 0$



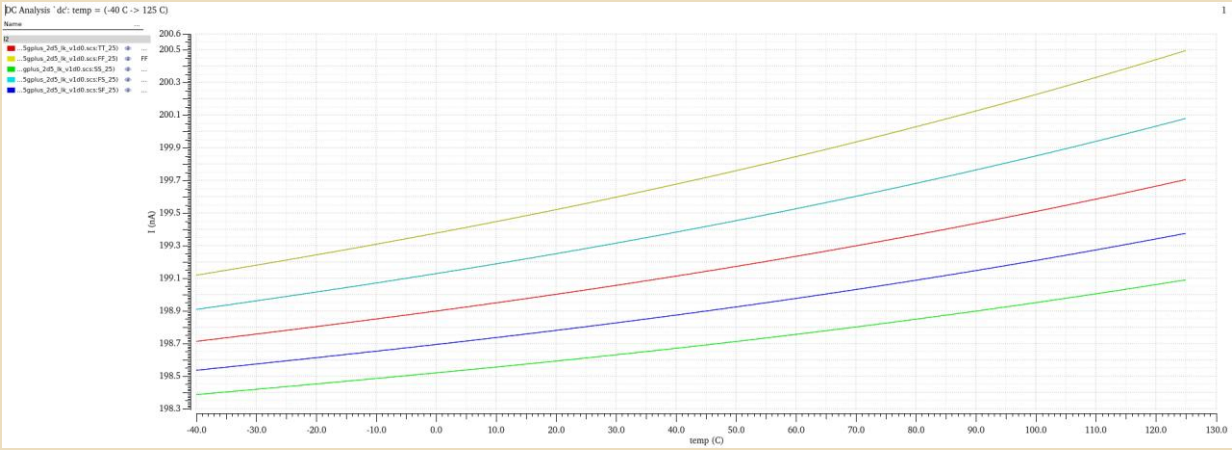
OP simulation



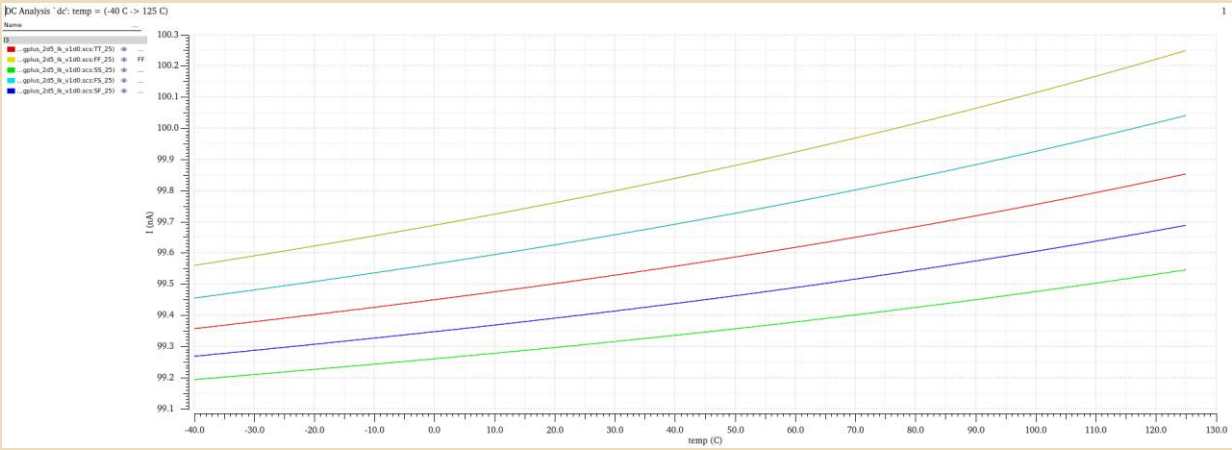
Ideal $I_B = 400nA$



I_1 variations



I_2 variations



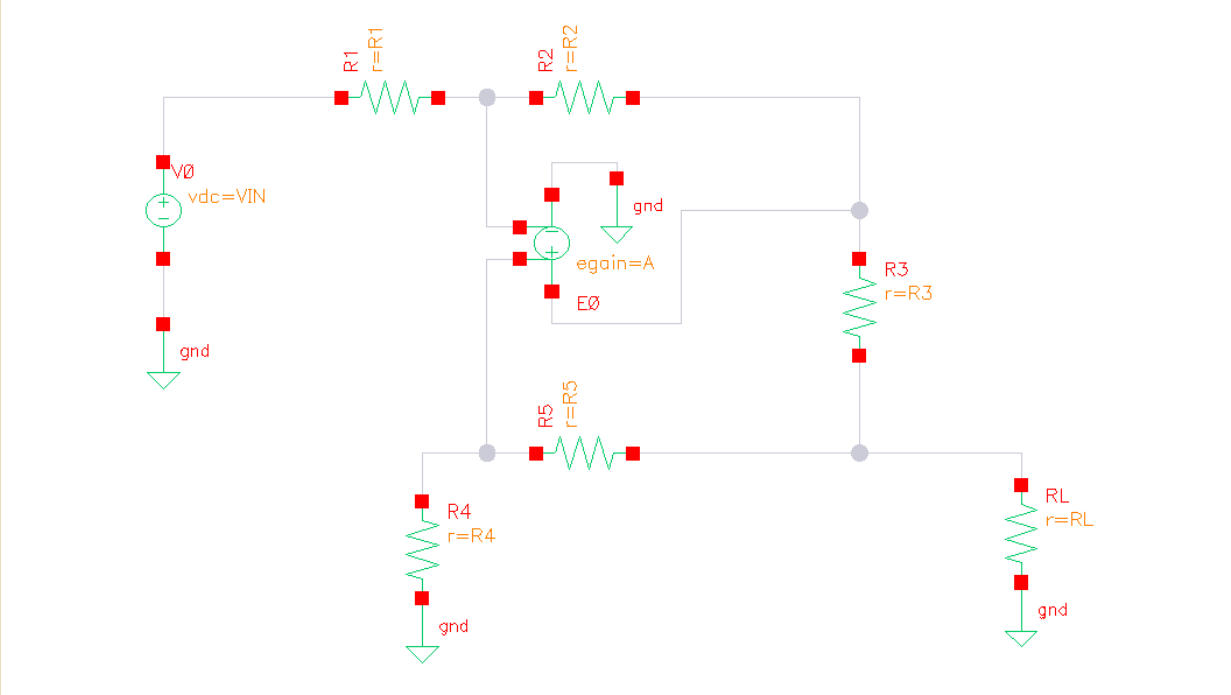
I_3 variations

Test	Output	Spec	Weight	Pass/Fail	Min	Max	TT	FF	SS	FS	SF
si-Bgenerator:1-CM-test:1	IB										
si-Bgenerator:1-CM-test:1	IB-variations				0	0	0	0	0	0	0
si-Bgenerator:1-CM-test:1	I1										
si-Bgenerator:1-CM-test:1	I1-variations				1.409n	2.757n	1.985n	2.757n	1.409n	2.343n	1.681n
si-Bgenerator:1-CM-test:1	I2										
si-Bgenerator:1-CM-test:1	I2-variations				704.5p	1.379n	992.7p	1.379n	704.5p	1.172n	840.4p
si-Bgenerator:1-CM-test:1	I3										
si-Bgenerator:1-CM-test:1	I3-variations				352.3p	689.3p	496.3p	689.3p	352.3p	585.8p	420.2p

The corner variations for I_3 is very good for ideal input

Voltage-to-Current Converter Design

Now we will design the voltage to current op amp, $V_{in} = 1V$ from BGR

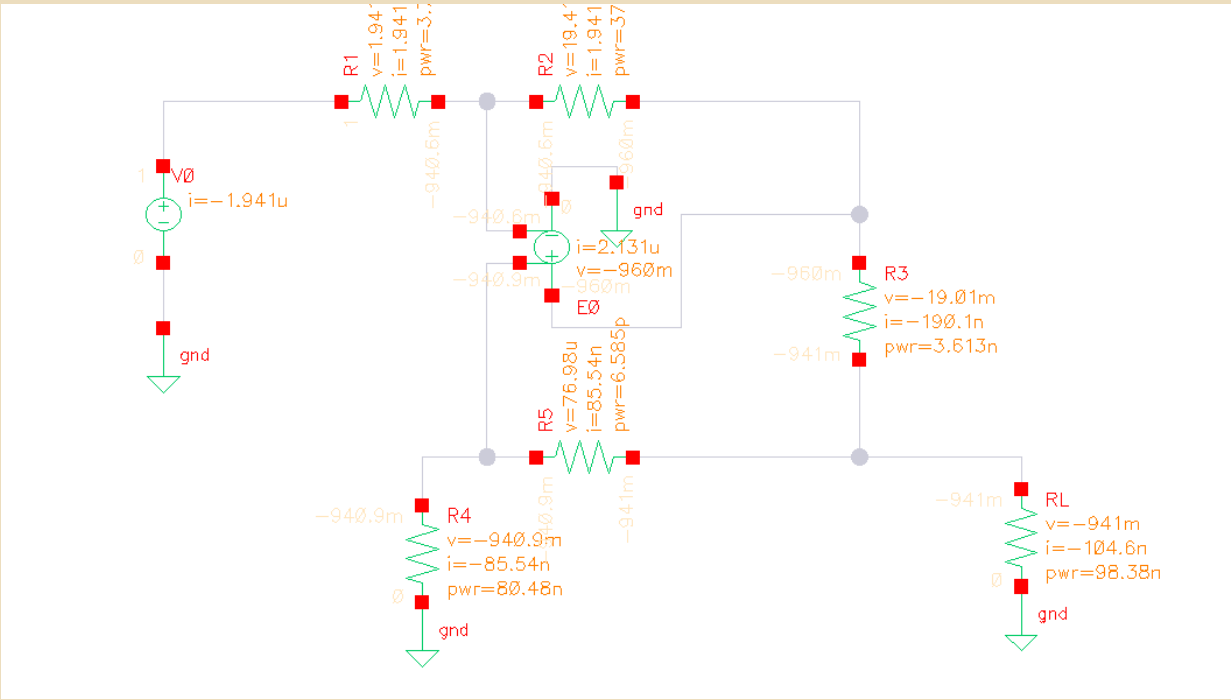


Using behavioral error amplifier with open loop gain $A = 3000$.

For $100nA$ output current, $I_{out} = -\frac{R_2}{R_1 R_3} V_{in} \Rightarrow \frac{R_2}{R_1 R_3} = 100 \times 10^{-9}$, let $R_2 = 10K\Omega$, $R_1 = 1M\Omega \Rightarrow R_3 = 100K\Omega$

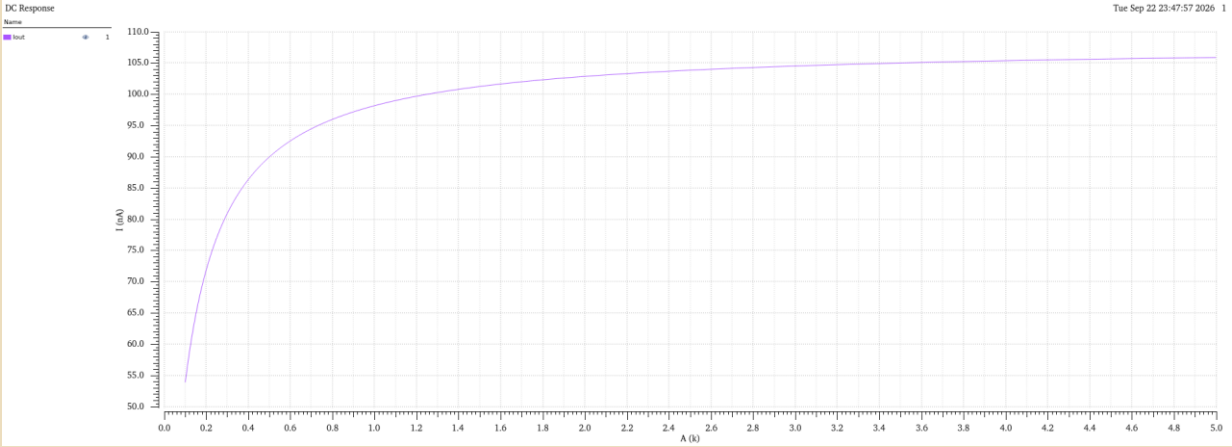
$R_1(R_3 + R_5) = R_2 R_4$, let $R_5 = 10K\Omega \Rightarrow R_4 = 11M\Omega$

$R_{L_{max}} = \frac{(R_4+R_3)\left(\frac{V_{sat}}{I_{outm}} - R_3\right)}{R_3+R_4+R_5}$, set $I_{outm} = 100nA \Rightarrow R_{L_{max}} \approx 9.81M\Omega$, set $R_L = 9M\Omega$



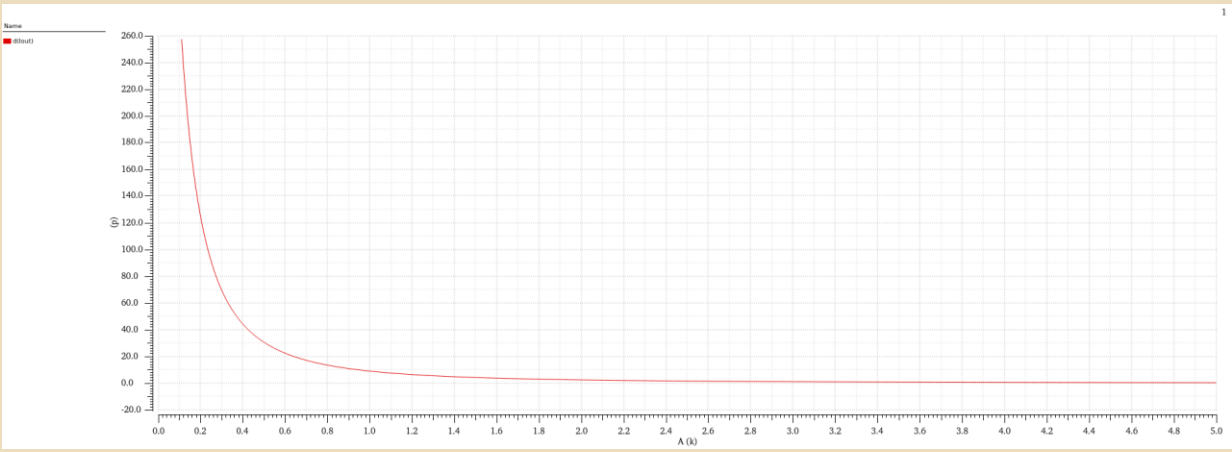
OP simulation

The output current $= 104.6nA$

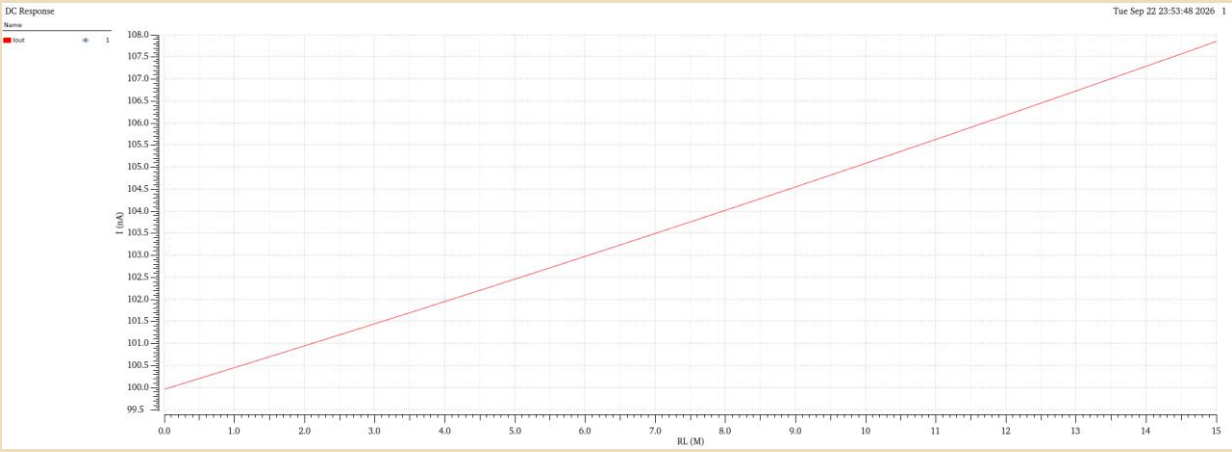


I_{out} *vs* A

The current almost saturate for high open loop gain

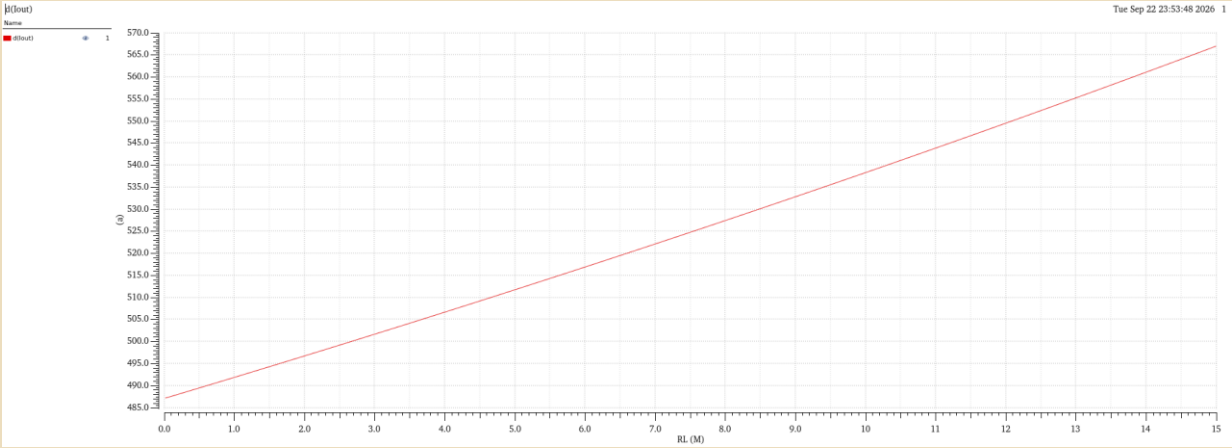


dI_{out}/dA



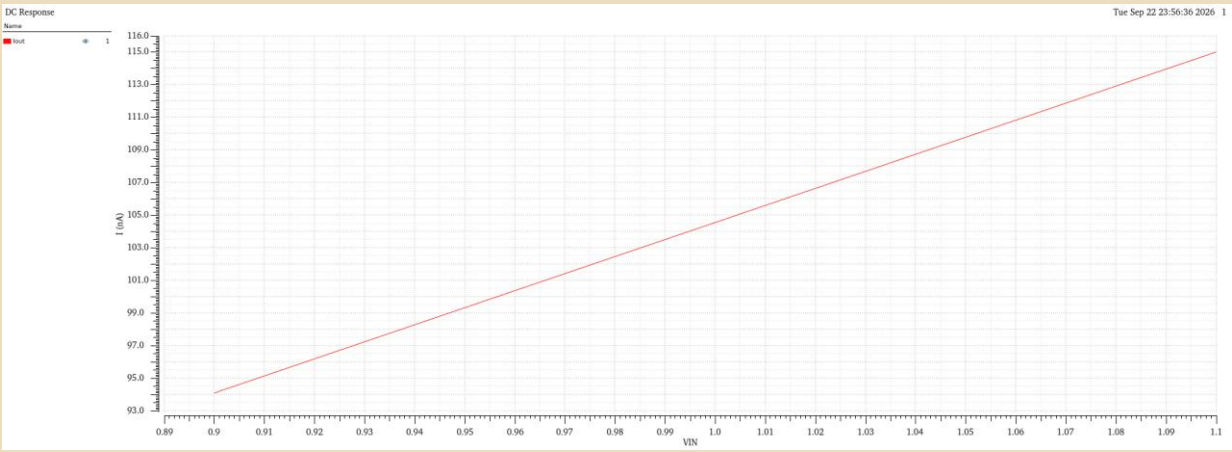
I_{out} *vs* R_L

As the load resistance increases, the current well

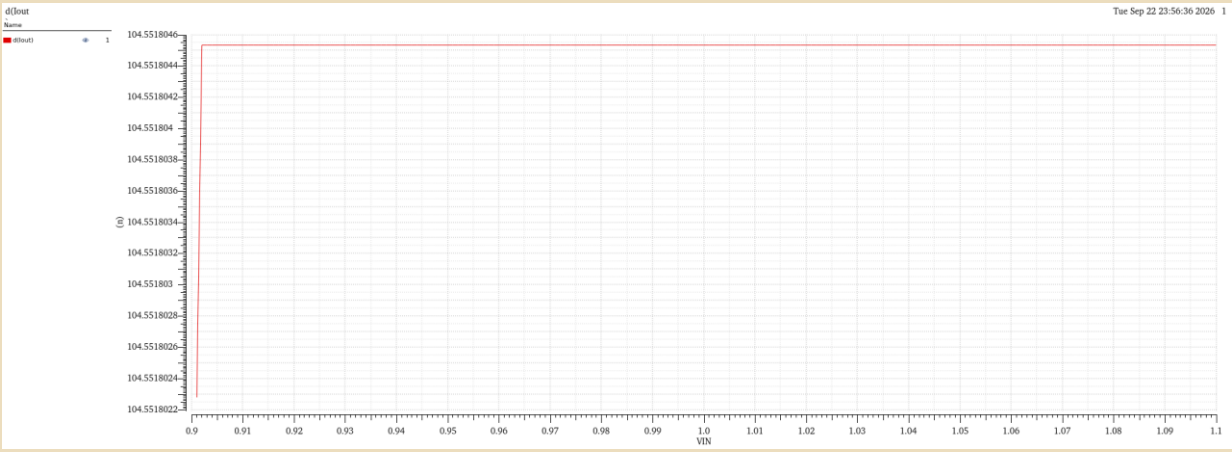


dI_{out}/dR_L

The increasing of the current with the load not exactly linear



$I_{out}\ vs\ V_{in}$

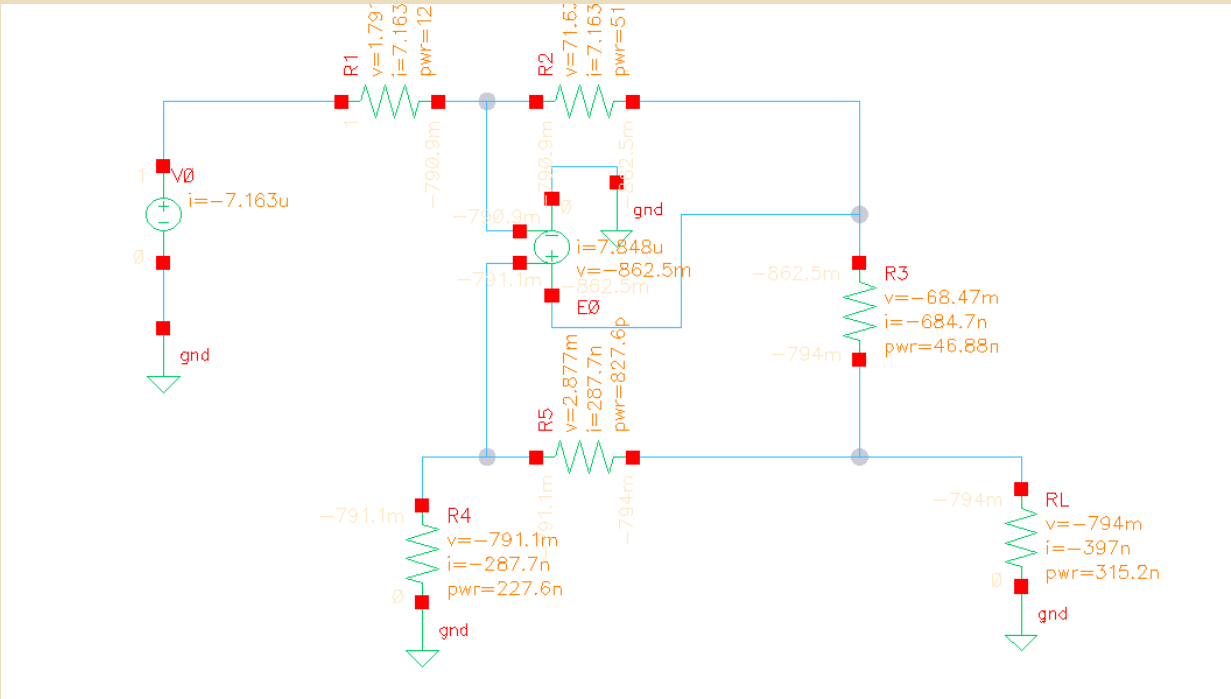


dI_{out}/dV_{in}

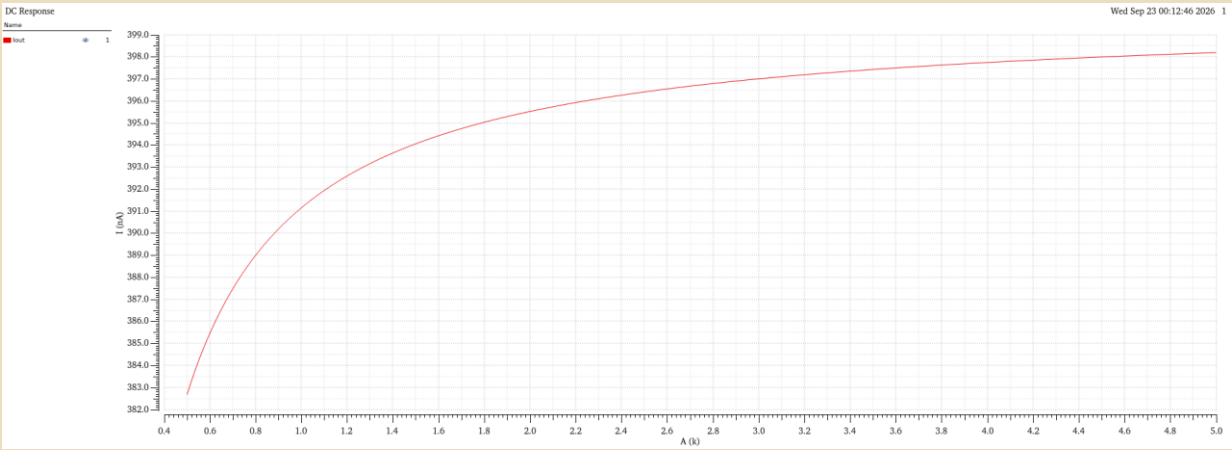
For $400nA$ output current, $I_{out} = -\frac{R_2}{R_1 R_3} V_{in} \Rightarrow \frac{R_2}{R_1 R_3} = 400 \times 10^{-9}$, let $R_2 = 10K\Omega$, $R_1 = 250K\Omega \Rightarrow R_3 = 100K\Omega$

$R_1(R_3 + R_5) = R_2 R_4$, let $R_5 = 10K\Omega \Rightarrow R_4 = 2.75M\Omega$

$R_{L_{max}} = \frac{(R_4+R_3)\left(\frac{V_{sat}}{I_{outm}} - R_3\right)}{R_3+R_4+R_5}$, set $I_{outm} = 400nA \Rightarrow R_{L_{max}} \approx 2.316M\Omega$, set $R_L = 2M\Omega$

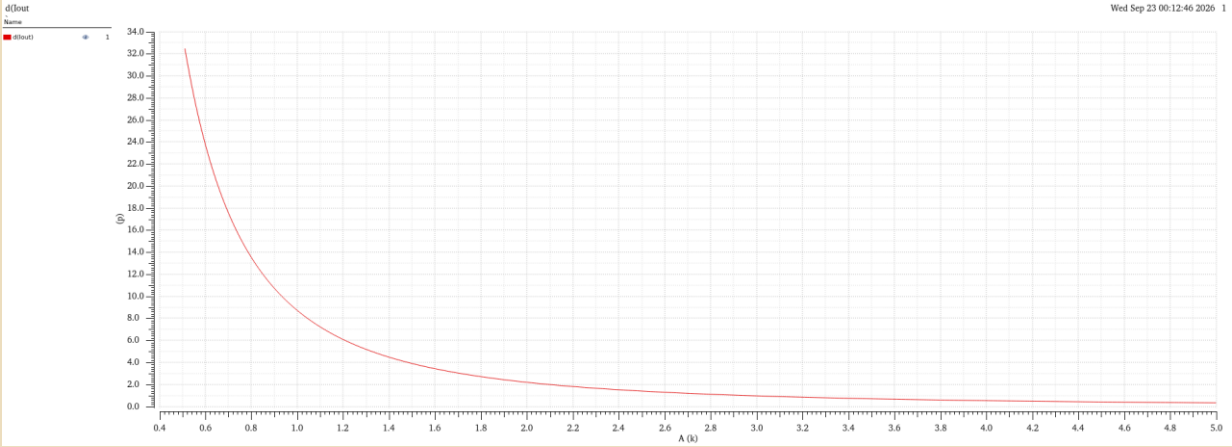


OP simulation

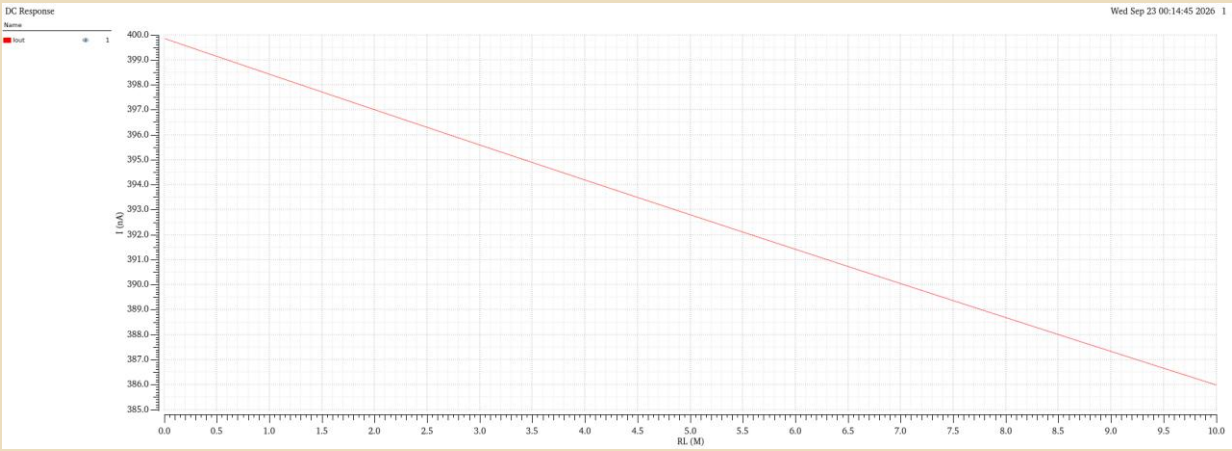


I_{out} *vs A*

The current almost saturate for high open loop gain

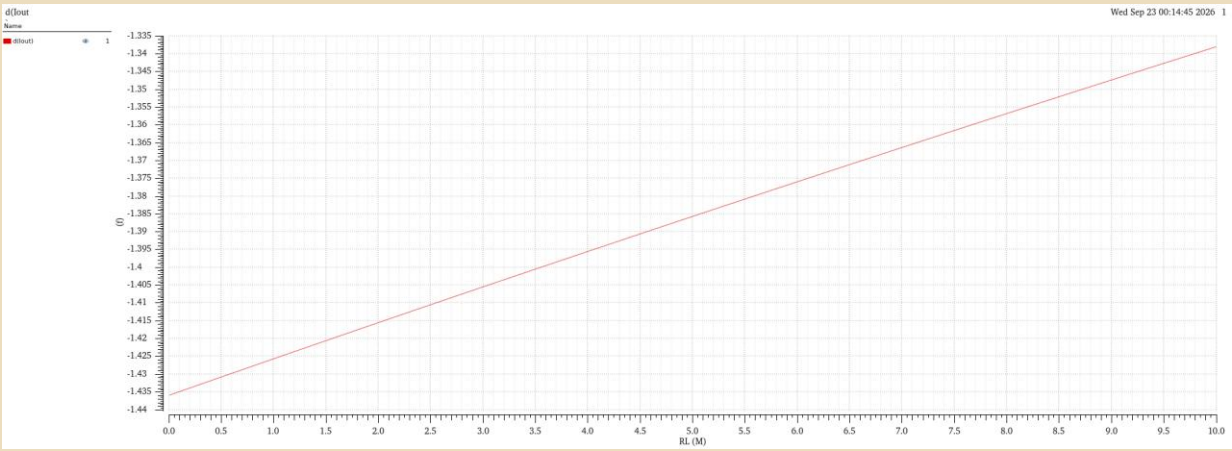


dI_{out}/d_A

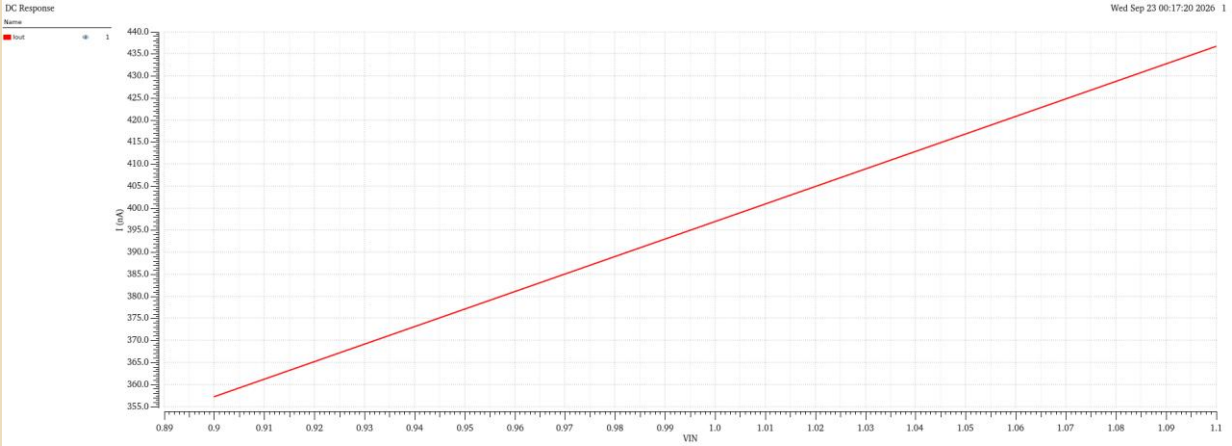


$I_{out}\ vs\ R_L$

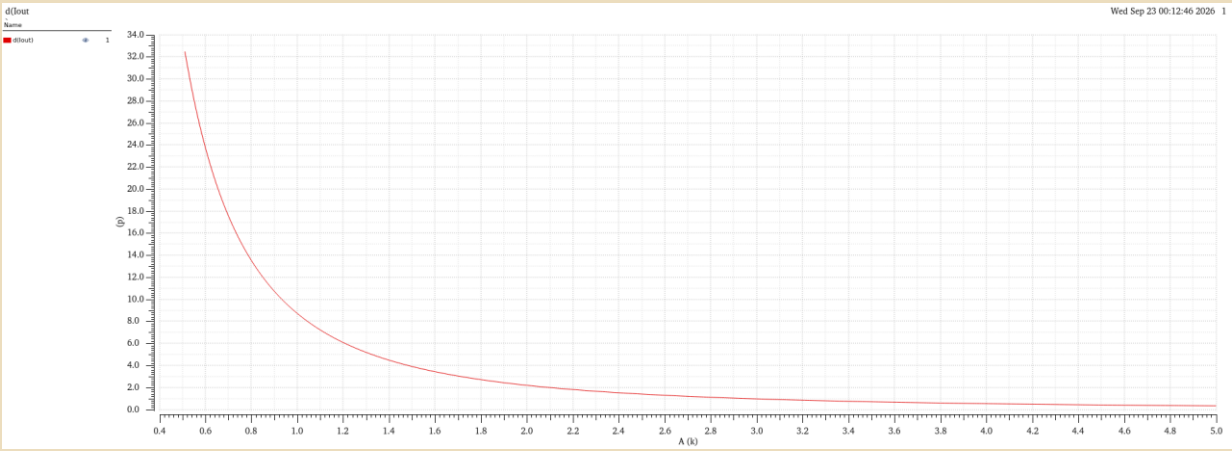
As the load resistance increases, the current decreases



dI_{out}/dR_L



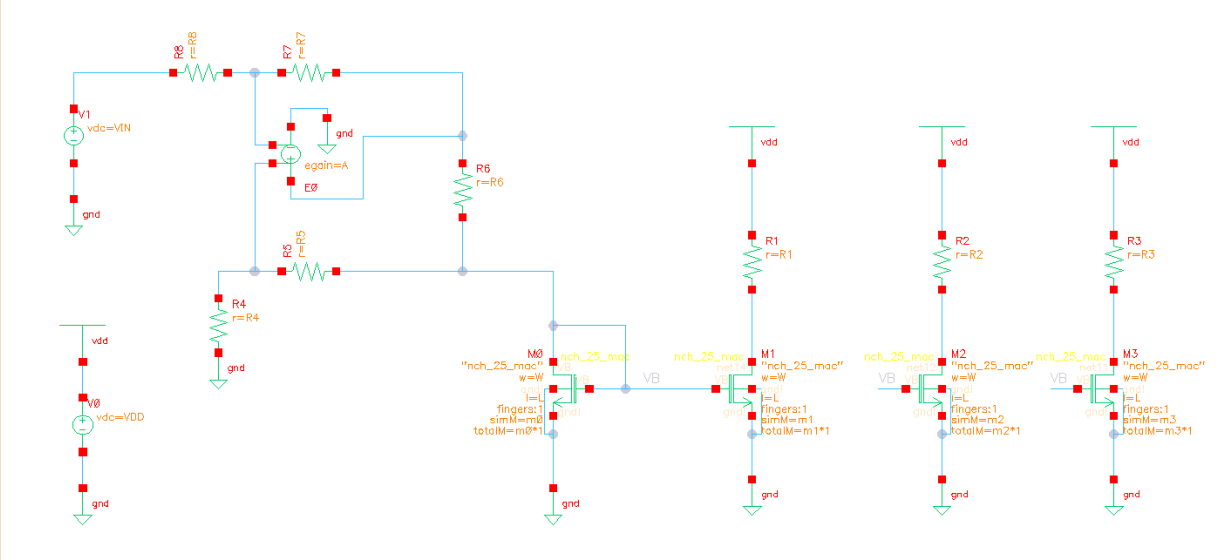
I_{out} *vs* V_{in}

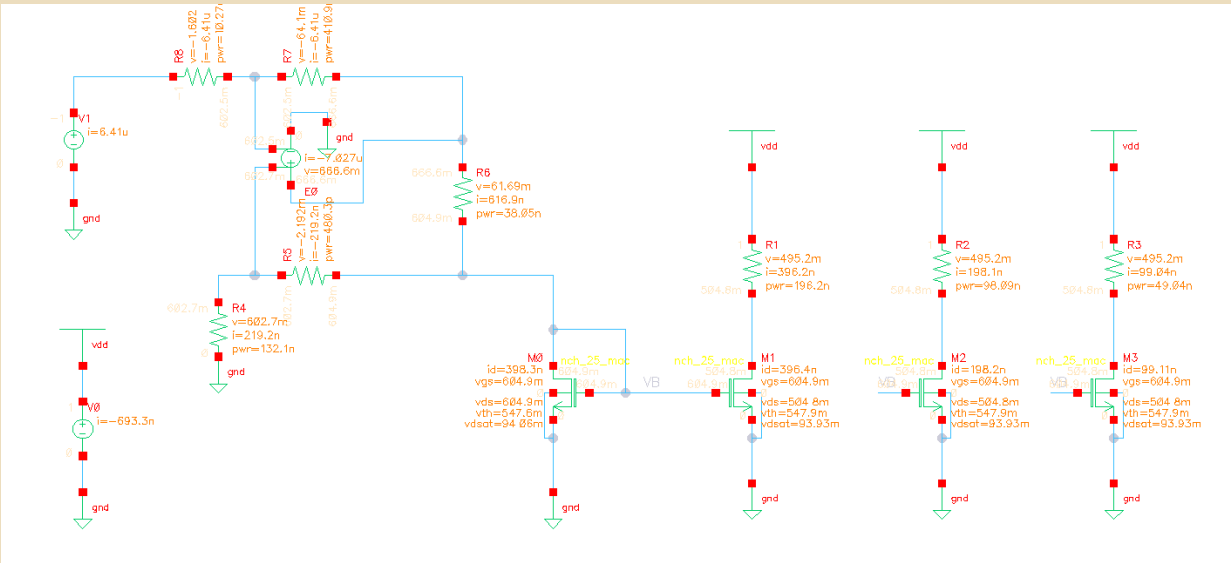


dI_{out}/dV_{in}

Combined V-I Converter & Current Mirror

After simulating the voltage to current converter and the current mirror, now we can combine them together



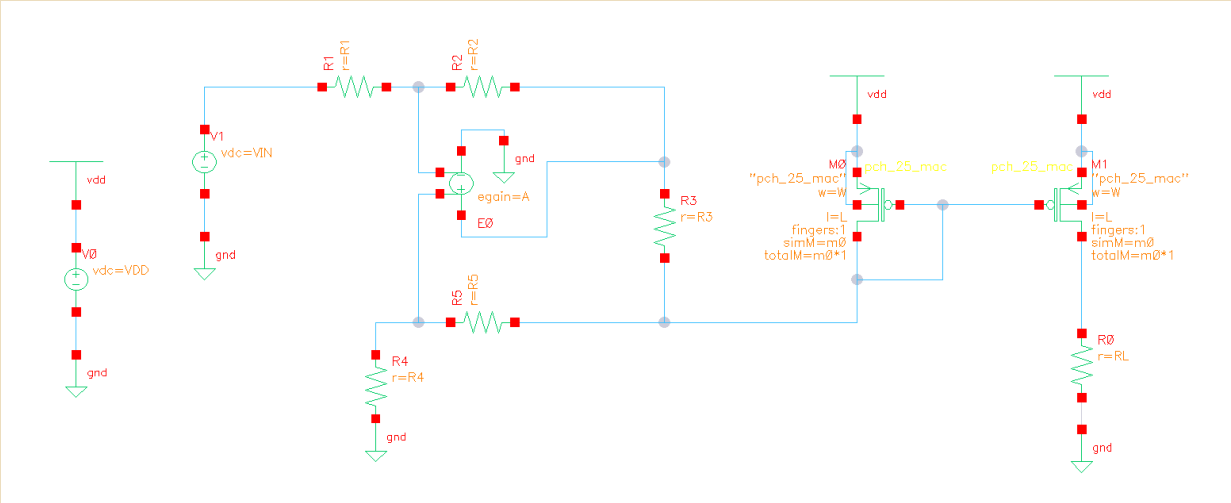


OP simulation

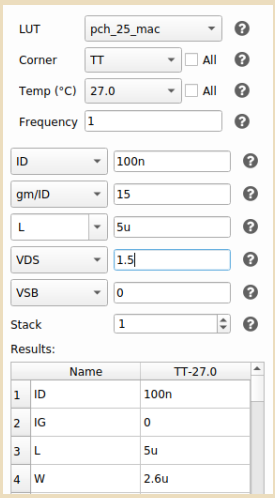
Notice that the current $I_3 = 99.11nA$ is much better than the current produced directly from the V-I converter $104.6nA$

PMOS-Based Current Mirror (Positive Input)

Note since the converter output current $I \propto -V_{in}$, i set the input voltage to be $-1V$, but for $+1V$ input voltage from BGR we can redesign the current mirror circuit by PMOS devices.



PMOS device sizing



set $V_{DD} = V_{bat} = 3V$ to avoid loading the BGR

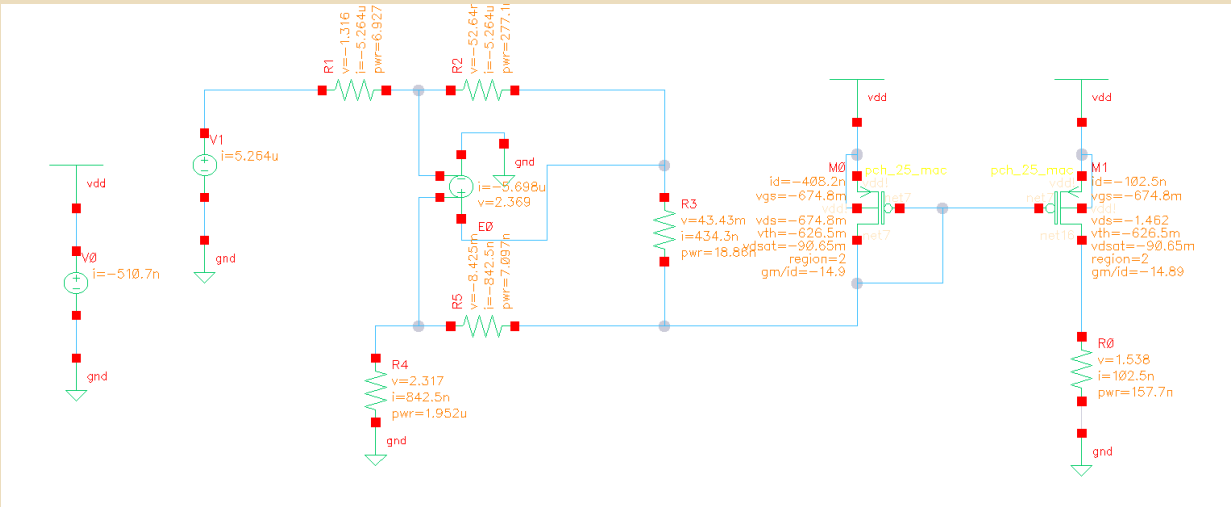
let $V_{DS} = 1.5V$, $g_m/I_D = 15$, max $L = 5\mu m$

From ADT, we find $W = 2.6\mu m$.

Setting mirroring ratios to get $100nA$ from $400nA$ output

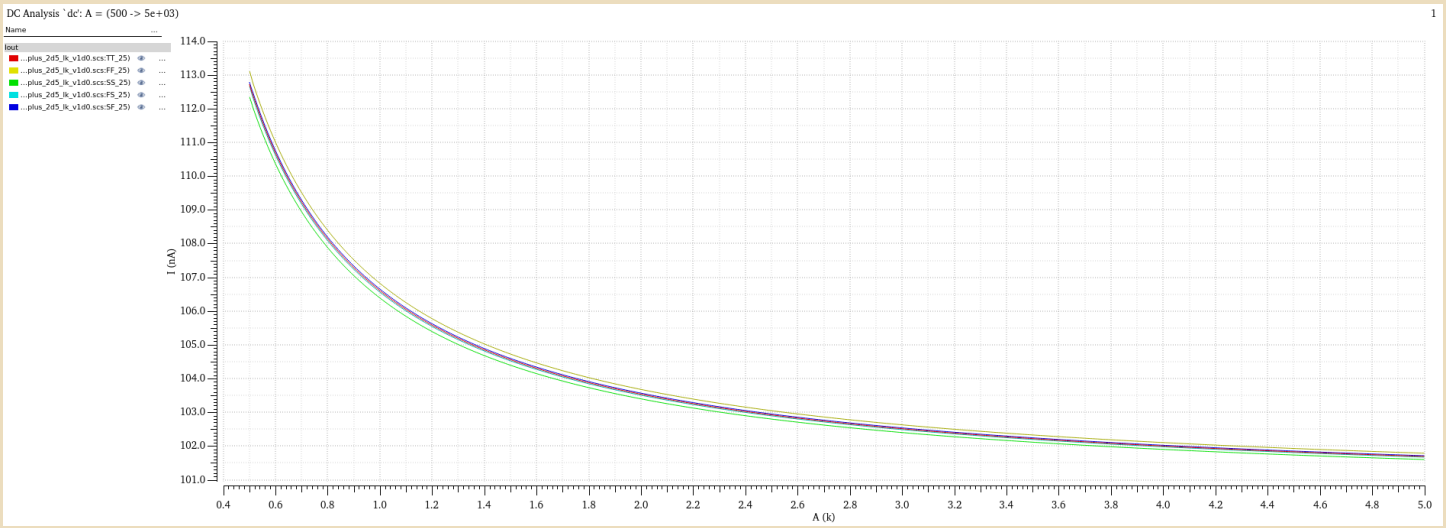
$\Rightarrow m_0 = 4, m_1 = 4$

Set $R_L = 1.5V/100nA = 15M\Omega$

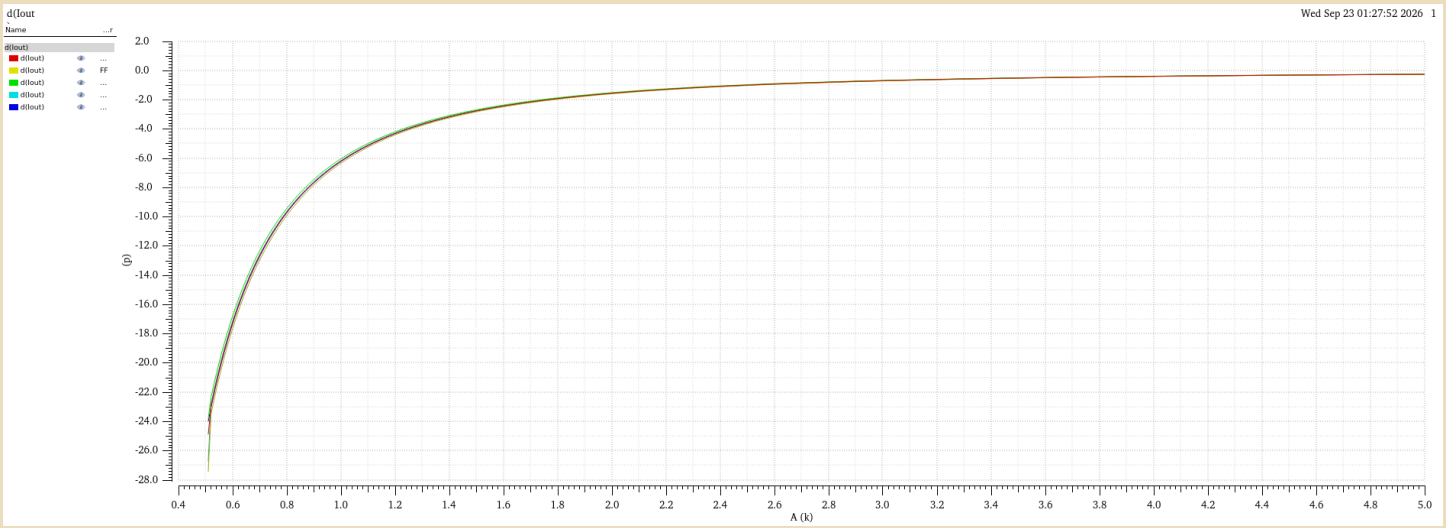


OP simulation

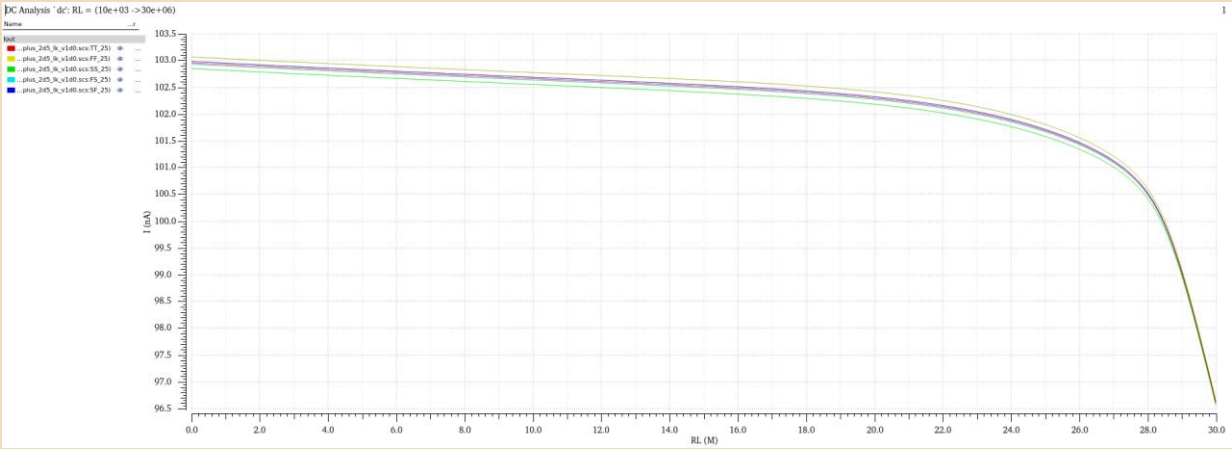
The output current $102.5nA$



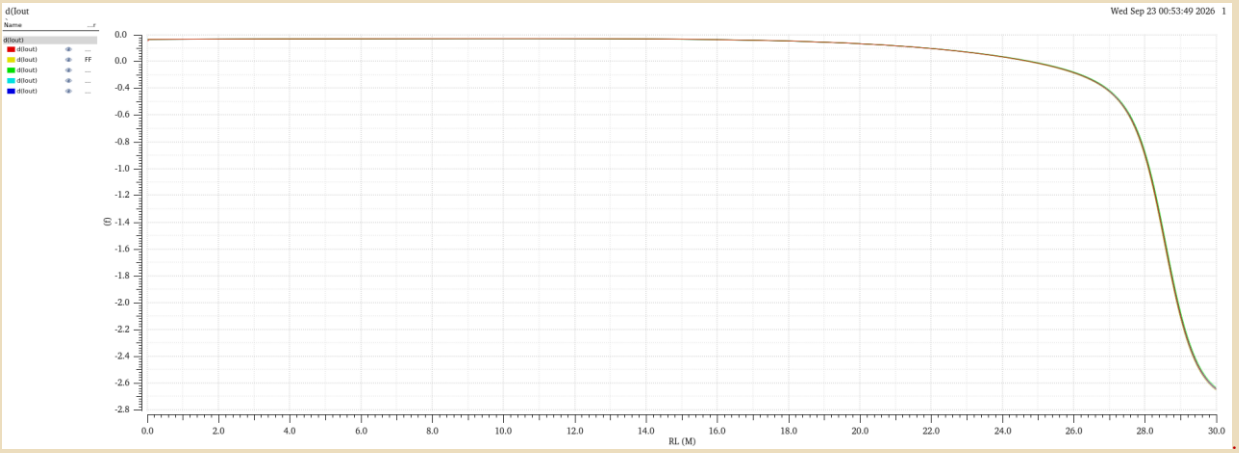
I_{out} *vs* A



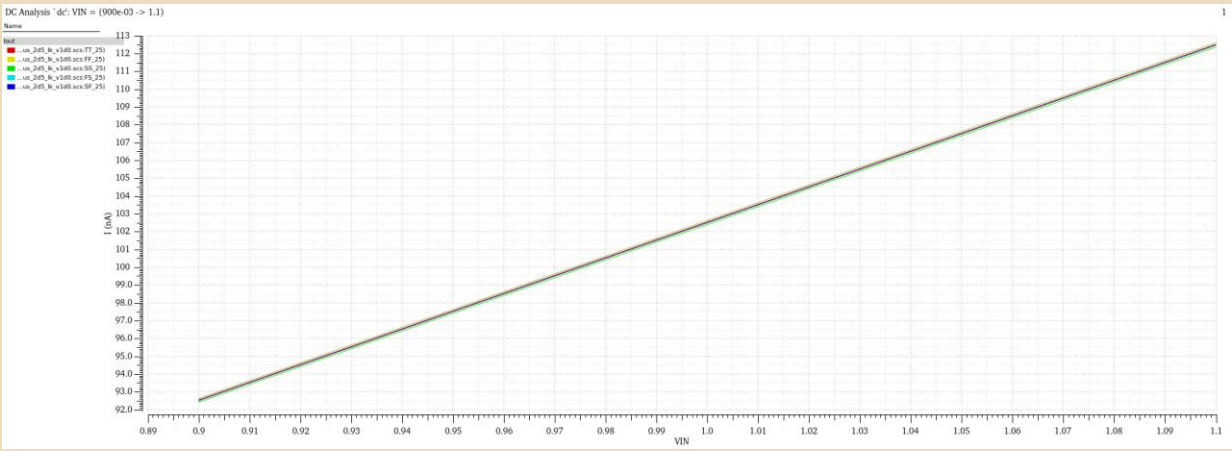
dI_{out}/d_A



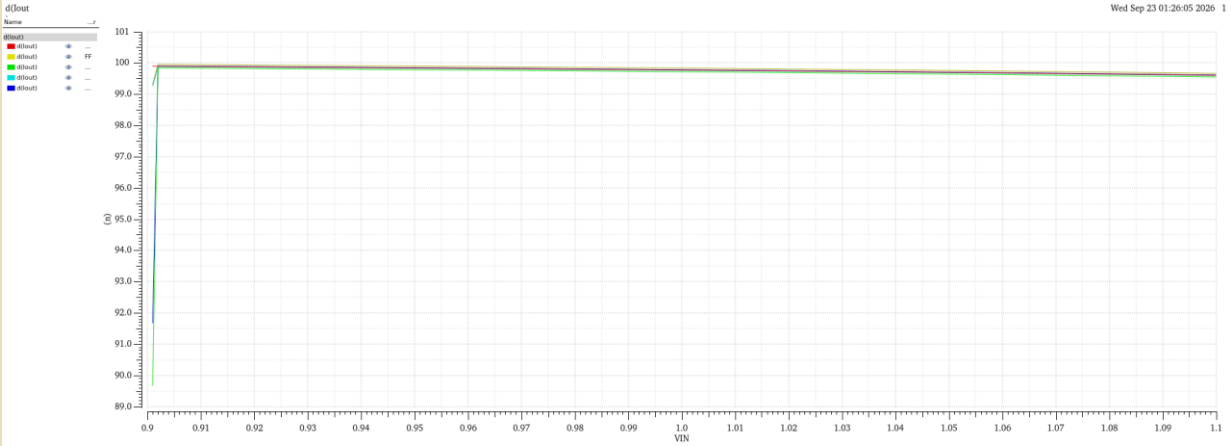
$I_{out}\ vs\ R_L$



dI_{out}/dR_L



$I_{out}\ vs\ V_{in}$


d(Iout
Wed Sep 23 01:26:05 2026 1
Name
d(Iout)
d(Iout)
d(Iout)
d(Iout)
d(Iout)
d(Iout)
FF
(a)
101
100
99.0
98.0
97.0
96.0
95.0
94.0
93.0
92.0
91.0
90.0
89.0
0.9
0.91
0.92
0.93
0.94
0.95
0.96
0.97
0.98
0.99
1.0
1.01
1.02
1.03
1.04
1.05
1.06
1.07
1.08
1.09
1.1
VIN

dI_{out}/dV_{in}

Power on Reset POR

1. Requirement Recap

The Power-On-Reset / Battery Monitor block must generate a reset signal that tracks the battery voltage against two thresholds, with hysteresis, while staying inside the PMU-wide ultra-low quiescent current budget.

Parameter	Target	Req #
Vth_high (rising, POR = '1')	≤ 2.5 V	R.034
Vth_low (falling, POR = '0')	≥ 2.3 V	R.035
Hysteresis (Vhyst)	≤ 200 mV	R.036
Current consumption (icon)	Budgeted from R.002 & R.009	R.033

2. Initial Architecture Exploration — Battery Supervisory Circuit (BSC)

The first architecture considered was the Battery Supervisory Circuit (BSC) described in the reference thesis (attached, “inhee_1.pdf”), reproduced below as Figure 1. It is a full-featured supervisory block built around six sub-blocks:

- A 23-device PMOS diode stack acting as a resistor-less voltage divider on VBAT, with feedback for hysteresis.
- A two-stage comparator (a continuous comparator followed by a clocked comparator) feeding an SR latch to produce COMPOUT / RESET.
- A delay generator (D flip-flop based) that produces the final RESETb after a controlled delay.
- A voltage reference stage (M1–M7) generating VREF and VBIAS for the comparator and divider.
- A low-battery-voltage protector that gates the clock generator via LBVRST.
- A leakage-based ring-oscillator clock generator supplying CLK to the clocked comparator.

This architecture is comprehensive and well suited to a supervisory IC with multi-die reset propagation, but it is significantly heavier than what the SiClash PMU specification (R.033–R.036) requires: it uses two comparator stages, a dedicated clock generator, a delay-locked reset chain, and a 23-device diode stack — all of which add current, area, and verification overhead beyond the project's single-threshold-pair, low-current target.

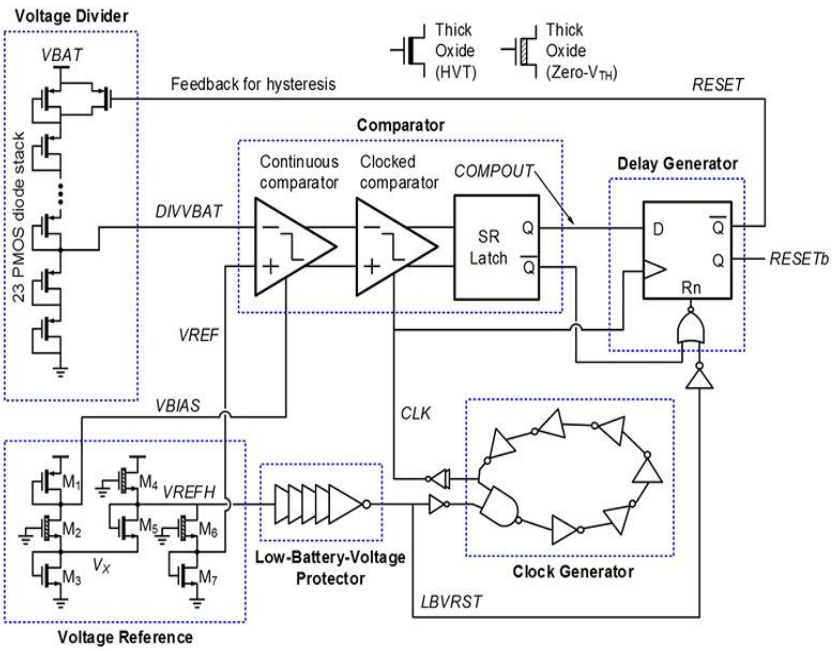


Figure 4.6: Circuit implementation of proposed BSC.

Battery Supervisory Circuit (BSC) architecture considered initially (source: reference thesis, Fig. 4.6).

3. Decision to Simplify

Given the project's current budget (Ipd ≤ 1 nA powerdown, icon budgeted from R.002/R.009) and the fact that only a single rising/falling threshold pair with hysteresis is required (no multi-die reset propagation, no externally clocked comparison), the BSC was judged to be over-specified for this design. The adaptation strategy below was used to decide what to keep, simplify, or remove:

Original BSC block	Adaptation applied for this project
Leakage oscillator	Remove if an external wake-up signal is available
Clocked comparator	Replace with a transiently enabled compact comparator or MOS detector
Long POR delay	Use a short, specification-based delay or system startup sequencing
Two-stage reference	Avoid if a sufficiently accurate MOS detector is possible
Low-battery protector	Retain only if needed for safe operation across the full Vbat range
Complex reset propagation	Remove if the PMU does not need multi-die reset propagation

Following this table, the design was re-targeted toward a simpler, resistor-less, continuously operating detector — the topology described in “An Ultra-Low Quiescent Current Resistor-Less Power on Reset Circuit” (IEEE Xplore, doc

#9141362), which matches the project's nA-level current budget without needing a clock generator or a two-stage clocked comparator.

4. Implemented Architecture

The circuit actually implemented and simulated in Virtuoso follows the resistor-less POR reference above, split into two blocks: a Power-On-Reset (POR) detector and a Brown-Out Detection (BOD) detector, each built from diode-connected PMOS/NMOS current-mirror legs biased from VDD, followed by inverter chains and XOR-based edge/pulse shaping to generate clean POR_PULSE / BOD_PULSE outputs and the POR1/POR2, BOD1/BOD2 status signals.

4.1 Reference circuit targeted for implementation

Figure 2 below is the exact circuit from the reference paper that this checkpoint set out to implement in Virtuoso. It reproduces the POR circuit, BOD circuit, and output circuit from the paper's Fig. 4, and is the direct source for the device naming (P1–P12, N1–N13) and signal names (POR1, POR2, BOD1, BOD2, POR_pulse, BOD_pulse, V1, V2, Vout) used in the schematics that follow.

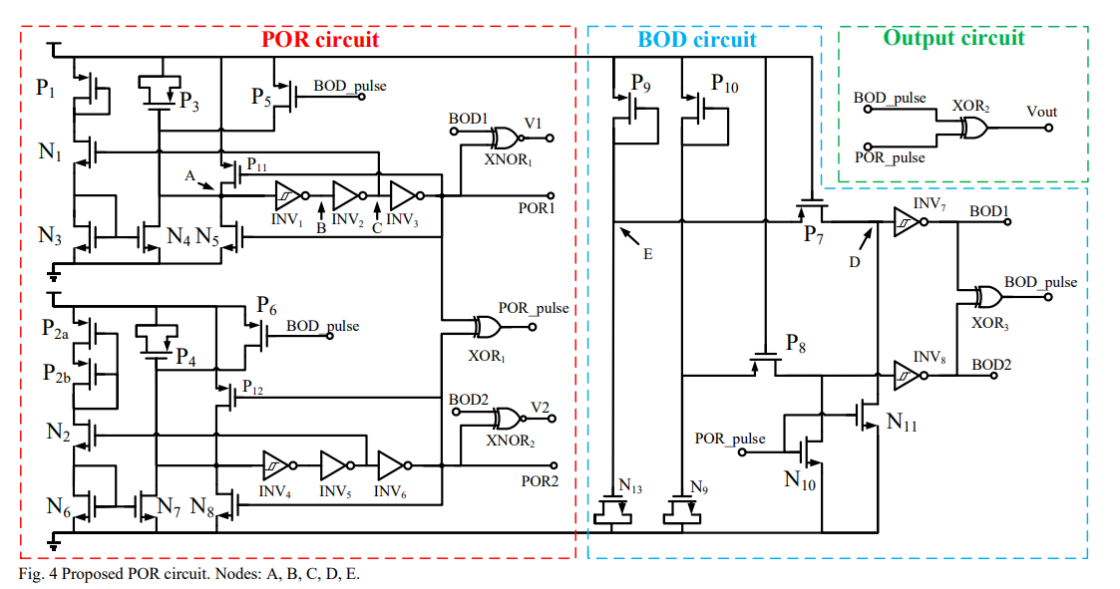
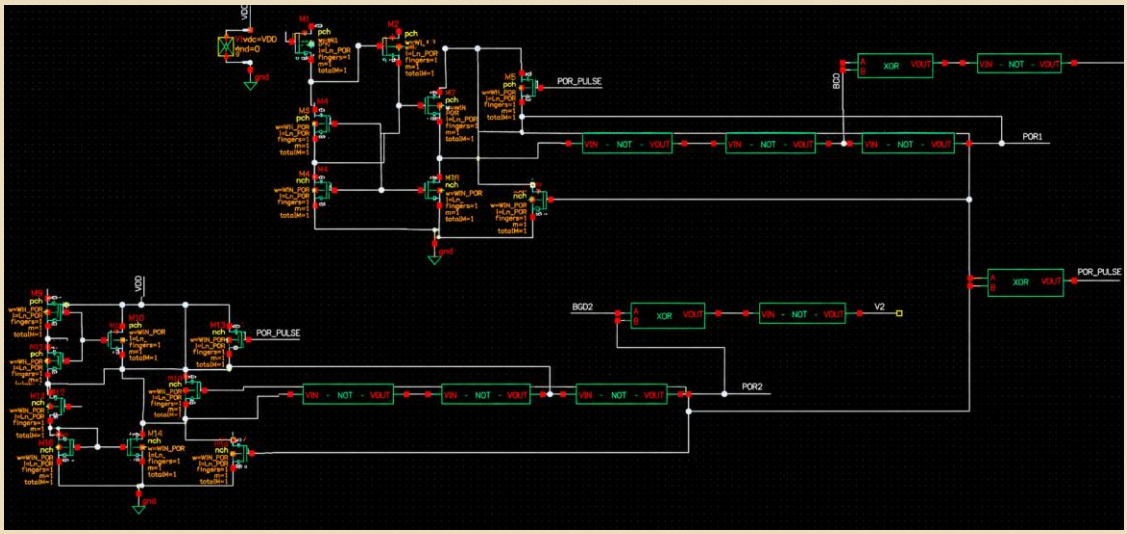


Fig. 4 Proposed POR circuit. Nodes: A, B, C, D, E.

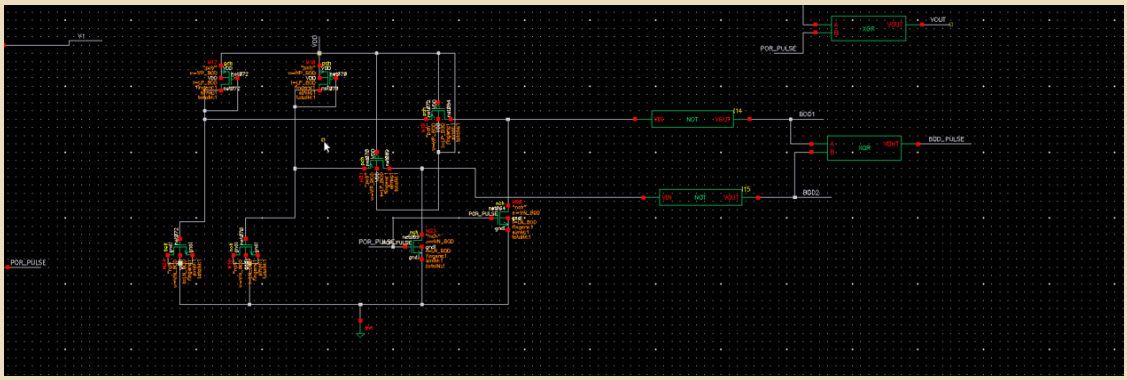
Reference POR/BOD/output circuit from "An Ultra-Low Quiescent Current Resistor-Less Power on Reset Circuit", IEEE Xplore, doc #9141362 [4]. Nodes A, B, C, D, E as labeled in the original paper.

4.2 Power-On-Reset (POR) schematic (as implemented in Virtuoso)



Implemented POR detector in Virtuoso: diode-connected PMOS/NMOS legs (M1–M16) generate POR1/POR2 and POR_PULSE through inverter chains and XOR pulse shaping.

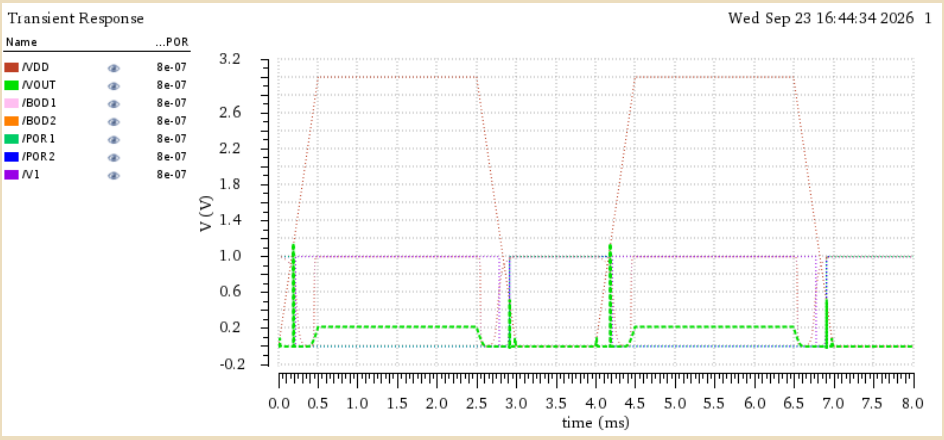
4.3 Brown-Out Detection (BOD) schematic (as implemented in Virtuoso)



Implemented BOD detector in Virtuoso, sharing the POR_PULSE signal and generating BOD1/BOD2 and BOD_PULSE.

5. Simulation Results and Open Issue

A transient testbench was run sweeping the battery/supply node up through both thresholds and back down. The resulting waveform is shown in Figure 5.



Transient response of the implemented POR/BOD chain: VDD (red, dotted) ramps toward ~3.2 V while VOUT (green) and the POR/BOD status lines (pink, orange, teal, blue) respond.

Current status: the two required trip thresholds (Vth_high ≈ 2.5 V rising, Vth_low ≥ 2.3 V falling, R.034/R.035) have not yet been met by the implemented diode-stack sizing. In the transient above, the POR/BOD status lines settle near 1.0 V and VOUT toggles between roughly 0 V and 1.05 V rather than tripping cleanly at the two target battery voltages, and the hysteresis window is not yet confirmed to sit within the 200 mV budget (R.036).

6. Root Cause and Next Steps

The most likely cause is that the diode-connected stack sizing (W/L of the PMOS/NMOS legs M1–M16) was carried over from an initial estimate rather than re-derived for the specific VDD-referenced thresholds needed here, since the reference IEEE topology was adapted from a different supply range and bias current than this project's VDD/VBAT budget. Planned next steps:

- Re-extract the subthreshold parameters (I_0, n, Vth0) for the actual PDK devices used, from a DC sweep of a single diode-connected leg.
- Re-derive the number of stacked devices and their W/L so the divider tap crosses 2.5 V on the rising edge and 2.3 V on the falling edge at the budgeted bias current (R.033).
- Re-run the PVT and Monte Carlo sweeps once the nominal thresholds are met, per the Checkpoint 5 verification plan.
- Re-check the hysteresis feedback path sizing against the 200 mV budget (R.036) once the primary thresholds are corrected.
- Revisit whether 65 nm was the right technology choice for this block: the reference circuit's diode-connected legs rely on subthreshold VGS drops to set the 2.3–2.5 V thresholds, and 65 nm's higher leakage and lower per-device threshold voltage may be part of why the divider is not landing on the target trip points. Compare against re-implementing at 130–180 nm, where longer-channel, higher-Vth devices give more predictable subthreshold VGS and easier threshold placement, before committing further sizing iterations at 65 nm.

References

[1] Analog Devices – “Powering ICs On and Off”, Analog Dialogue.

[2] IEICE Electronics Express, vol. 19, no. 10, article 19.20220119.

[3] Authorea preprint, doi: 10.22541/au.167426831.19406276/v1.

[4] “An Ultra-Low Quiescent Current Resistor-Less Power on Reset Circuit”, IEEE Xplore, doc #9141362 — topology implemented in Virtuoso for this checkpoint.

DFT

The DFT strategy for the PMU is still under investigation. The literature reviewed so far, "Low Power Testing: A Study of Power Aware Test Insertion Techniques" (Yawalkar and Gupta, CIPHER 2026), addresses digital scan-based DFT, so it is used only as background on test-power overhead and isolation between power domains. The analog sense/force scheme through the ATB is still to be defined and verified.